MULTILABEL NEWS ARTICLE CLASSIFICATION USING NLP

A project submitted to the Bharathidasan
University in partial fulfillment of the requirements for
the award of the Degree of

MASTER OF SCIENCE IN DATA SCIENCE

Submitted by
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Under the guidance of
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BISHOP HEBER COLLEGE (AUTONOMOUS)
"Nationally Reaccredited at A++ Grade with a CGPA of 3.69 out of 4 in NAAC IV Cycle"
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(Affiliated to Bharathidasan University)
TIRUCHIRAPPALLI-620 017
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DECLARATION

I hereby declare that the project work presented is originally done by me under the guidance of **Dr. C. LINDA HEPSIBA, M.Sc., M.Phil., Ph.D., NET., Assistant Professor (SS), Department of Data Science, Bishop Heber College (Autonomous), Affiliated to Bharathidasan University**, **Tiruchirappalli-620 017** and has not been included in any other project submitted for any other degree.

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DEPARTMENT OF DATA SCIENCE

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PAVITHRA C

ABSTRACT

The emergence of online news platforms has led to a large amount of unstructured text data, making it difficult to automate news categorization. In practical applications, news articles are categorized into multiple categories at the same time, which cannot be processed by traditional single-label classification methods. To overcome this difficulty, the project proposes a Multi-Label News Article Classification system based on BERT with Headline, Summary, and Body Fusion. The proposed system uses the Reuters-21578 dataset, where each news article is divided into three parts: headline, summary, and body. Each part is processed separately by a Bidirectional Encoder Representations from Transformers (BERT) model to extract contextual features. The features are then fused by a concatenation-based fusion layer, allowing the model to learn high-level and low-level semantic information. The sigmoid activation function is used in the output layer to enable multi-label classification. The performance of the system is measured using conventional multi-label metrics like precision, recall, and F1-score. The experimental outcome shows that the proposed fusion-based BERT model performs better than conventional machine learning techniques and single-input deep learning models. Moreover, the trained model is deployed using Streamlit, which offers a friendly interface for real-time news classification. The outcome shows that the proposed model is efficient, scalable, and applicable for real-world multi-label news classification tasks.

TABLE OF CONTENTS

245229137

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II M.Sc Project

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1. INTRODUCTION

Digital news platform and online news media has created a situation whereby the number of news items that are being created every day is increasing beyond anything that has been witnessed before. The production of news content is no longer limited to the conventional media platforms but it is being generated on an unrelenting basis using web portals, mobile applications and social media platforms. The huge increase in textual data has rendered it very cumbersome and inefficient to subjectively categorize news items into the specific category. Automated news classification systems have therefore become a constituent component of the contemporary information search systems and recommender systems. But in real life situations, news articles are likely to fall into more than one classification at a time, therefore, making multi-label classification a more practical and challenging task as compared to single-label classification, which is more conventional.

Traditional machine learning models like Naive Bayes, Support vectors and decision trees are extremely reliant on hand-crafted features and basic text representations. Such algorithms fail to comprehend deep contextual semantics particularly where complex news articles involving a lot of words are involved. Moreover, the majority of conventional algorithms assume that news articles are one continuous piece of text ignoring the natural structural parts of news articles like the title, summary, and body. All these elements possess their specific semantic values; to mention a few the title offers a summary, summary offers highlights, and body offers the contextual information.

Even though BERT proves highly effective in tasks involving the classification of the text, a lot of the current research studies continue to rely on BERT on news article as the input sequence and does not address the structural division of the article. This renders the model less able to acquire various semantic patterns of various sections of the text. To address this weakness the proposed system suggests a fusion based architecture in which the headline, summary and body of the news article are processed individually by applying separate BERT encoders. The resultant representations of features are then fused with the help of a fusion layer where the model is free to learn the complementary semantic information of all its components and produce more full-fledged document representation.

The suggested Multi-Label News Article Classification System with the combination of Headline, Summary, and Body Fusion based on BERT is compiled to categorize news items into various pertinent labels successfully in cases of the Reuters-21578 dataset. This system is a multi-label classification system that relies on a sigmoid activation function and measures performance in terms of precision, recall, and F1-score. Additionally, the trained model is implemented with the help of Streamlit that provides an interactive and convenient web API. The proposed system is practically applicable, and its applicability in the real-world news analytics, recommendation, and intelligent content filtering tasks is presented.

Multi-label news classification is also complicated by the fact that similar categories overlap with each other and label distributions are uneven in large datasets like Reuters-21578. An article in the news can be both economic, trade, political, and financial, meaning that the classification system must independently but semantically consistent predict each label. The traditional classification organization is likely to fail on these occasions due to its failure to reflect the interrelations and intersections among the labels. This necessitates the use of advanced deep learning designs that are able to effectively process many labels and still be semantically consistent.

Long-form text content is another significant problem in the classification of news articles. In comparison, when the news is presented, news articles are more likely to exceed the usual input length limit of transformer models and therefore the information could be lost to the crucial details of the story. The proposed system will extract and process the important information of each of the parts of the news article by dividing the news articles into their headline, summary and body part. This approach to processing, besides contributing to the enhanced extraction of the contextual information, contributes to enhancing the computational efficiency.

Besides, the relevance of the proposed system is supported by the fact that the model deployment process is embedded into the framework of the project in question. The proposed system, by using the Streamlit implementation of the trained BERT-based fusion system, does not only hold in the theoretical experimentation but also turns out to be operational in the practical environment. The proposed system can be used in the context of news aggregation platforms, automated tagging systems, and intelligent content recommendation engines by the web interface of the proposed system allowing users to provide dynamic news content and the system offers instantaneous multi-label predictions.

1.1 Background of the study

The increasing reliance on digital information systems has resulted in a significant change in how news is consumed, processed and disseminated. The users of the current generation require news that is relevant and timely and this has put enormous strain on the information management systems. The automated classification systems play an important role in enabling effective indexing, retrieval and recommendation of news articles. Without the creation of intelligent classification systems, it will be impossible to manage the news repositories on a mass scale leading to information overload and reduced user interaction.

The other major problem in classifying news articles is that the news categories are dynamic. Some categories of news such as finance, politics, technology, and international issues are not rigid and at times such news have a tendency of intersecting. As an example, a policy article about the international trade can be both economic and political at the same time. This partiality of categories renders the classification of the news articles in set categories impossible and futile. There is a need to have multi-label classification models to reflect the real-life semantic complexity of news articles.

Reuters-21578 is one popular dataset that has been employed as a standard dataset to test the performance of the text classification models due to the complex category structure and practical applications. The dataset contains a large pool of news articles pertaining to different categories, and this renders it appropriate in testing the efficacy of the models in both single-label classification and also in multi-label classification. However, to utilize this dataset to the maximum, one will be required to employ complex modelling methods that are capable of handling such concerns as the class imbalance, the co-occurring of labels, and document length. This also shows the significance of powerful representation learning models such as BERT.

Transformer architectures have been increasingly popular in the recent past as the most prominent solutions to natural language understanding problems. Specifically, BERT has demonstrated spectacular performance in terms of numerous NLP problems including text classification, question answering, and sentiment analysis. Combining the ability of the model to train on a large text corpus, but fine-tune on task-specific data allows the model to generalize effectively even with limited labeled data. This is an important property of BERT that is why it can be a perfect choice when it comes to complex tasks like the multi label classification of news articles.

There are however computational and representation difficulties with the use of BERT on news items. Articles containing news take more time than the maximum input length of the transformer model. The outcome of this is truncation of the article and the important contextual information lost. Moreover, the model considers the whole article as a chain. This undermines the significance of such important elements of the article as headlines and summaries. There is a need to be more organized in dealing with these challenges.

The fusion of features is an efficient approach that can merge the information of multiple representations or sources. Regarding the news articles, feature fusion gives an opportunity to combine semantic features which were independently extracted without references to headlines, summaries and bodies. With a combination of these representations, the model can be able to generalize high-level thematic knowledge and contextual fine-grained knowledge. Fusion-based architecture has been established to work effectively in enhancing classification accuracy particularly in the case of heterogeneous inputs.

The growing demand of web-based machine learning applications also contributes to the necessity of the deployment frameworks integration in research. The implementation of the classification models with the assistance of the interactive programs like Streamlit is useful in demonstrating how the models apply into real world. This assists in bridging the research and practical uses and users have the ability to interact with intelligent systems in real time. This end-to-end perspective contributes to the topicality of the study and effectiveness of NLP methods within the context of the real-life information systems.

The rapid development of the need of smart text classification solutions can also be linked to the fact that Unilever has been rapidly integrating the artificial intelligence technology into the information management system. Automated tagging and classification are very essential to the news industry, digital libraries and news aggregation sites to facilitate the other activities which include search engine optimization, trend analysis and news personalization. The accuracy of multi-label classification assists in ensuring that such systems label each piece of news according to different topics in which it is relevant and thereby make information more accessible. As the amount and rate of news data continue to grow, it is not only beneficial but also necessary to use deep learning-based classification structures in the present digital world.

1.2 Problem Statement

The high rate at which the online news content of the digital world increases has created a big problem of handling a huge quantity of unstructured text data. Different organizations such as international news bodies, financial organizations, government websites, and individual news channels generate news articles on a continuous basis. Manually classifying such a large volume of data is time-consuming and also very prone to errors. Thus, it is urgent to create automated tools that will be effective in classifying news articles.

In the real-life news publishing applications, news articles are more likely to have more than a single topic of concern. As an example, economic policies, political decisions and the impact of international trade can be covered in one news. The categories in the traditional classification processes are usually given to each document separately, which is not in any way representative of semantic nature of news articles.

The latest state-of-the-art automated news classification systems rely on the traditional machine learning approaches, whereby they utilize manual features such as bag-of-words or TF-IDF. Even though these algorithms are computationally effective, they do not allow capturing word-to-word and word-to-word dependencies, and semantic relations of a long text. As news articles are associated with complicated sentence structure and terms that belong to the domain of experience, shallow features results in the inadequate generalization and poor classification.

Although the deep learning models like the Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have made improvements, it still has a few limitations in applying it to news articles. This can be attributed to the fact that models like LSTMs are not good at long-range dependencies whereas CNN-based models concentrate on local information. In addition, the majority of deep learning models treat the whole news article as a single text block and do not take into account the form of headlines, summaries, and body text.

The second major problem in news classification is poor use of various textual parts of news article. Each of the headline, summary, and body present semantic information that is unique but all the current models do not utilize this structure. The failure to pay attention to these elements results in loss of information and the inability of the model to learn fine-grained representations. Models that can handle these components separately and as well as combine their semantic representations to enhance the accuracy of their classification are required.

An intelligent, scalable and contextual multi-label news classification system is required in order to combat these problems. Such a system needs to be capable of extracting very rich semantic interconnections, arbitrarily integrating information across different text constituents, and performing a set of correct multi-label predictions. This issue is resolved in this project, where a BERT based fusion model which incorporates headline, summary and body properties and applies sigmoid based multi label classification with good evaluation rates is created.

1.3 Objectives

The main objectives of this project are the following:

- Design and come up with an automated multi-label news article classification system which can categorize a news article into several tags.
- The Reuters-21578 dataset is a benchmark dataset that should be used to evaluate the performance of multi-label news classification.
- To analyze the news articles in detail, extract and preprocess news articles into headline, summary, and body parts.
- Prepare and derive BERT-based contextual feature extraction model to determine the semantic and syntactic relationship of the news articles.
- To create and implement a fusion-based architecture that is a concatenated representation of embeddings of the headline, summary, and body to enhance the overall semantic representation.
- To make use of a sigmoid activation function in the output layer to usefully perform multi-label prediction.
- To apply the model suggested on the typical performance indicators of precision, recall, and F1-score.
- To analyze how text fusion influences the accuracy of classification in comparison with models that receive single text inputs.

- To deal with the imbalance issue in labels, which is typically present in multi-label datasets in news classification.
- In order to make the proposed system scalable with the unstructured news information of large size.
- To build an effective training and validation of the deep learning-based text classification tasks.
- To deploy the trained classification model with the help of the Streamlit to develop an interactive and user-friendly web interface.
- To test the system by carrying out extensive testing, which would involve unit, integration testing and system testing stages.
- To support the development of a reproducible experimental setup with the help of standardized libraries and tools PyTorch and HuggingFace Transformers.

1.4 Scope of the Project

The focus of the given project is the creation of an automated multi-label system of news article classifier, utilizing the most recent methods of natural language processing. This system will be in a position of processing and classifying news articles to numerous relevant categories simultaneously as it happens with actual news articles. The scope will include the process as a whole and that is, data preprocessing through classification.

The dataset used in the current project is Reuters-21578 and includes a large collection of news articles in English language, encompassing many areas of knowledge, including economics, trade, politics, and industries. The text data available in the Reuters-21578 dataset is the only limit to the project and training and testing of the system is done by the category labels that are already present in the dataset. A well-established benchmark dataset is used in the project guaranteeing that the project is reproducible and comparable with the existing research work.

The organized news article processing is one of the main points of the project scope. Instead of treating an article as one piece of text, the system can work on the headline, summary and the body of the article individually. This is one of the major features of project scope since it allows the system to utilize the semantic information that exists in various sections of an article in order to enhance the accuracy of the classification process. In the project scope, the development of techniques that would allow the system to process articles of different sizes will be developed.

The design and implementation of BERT-based deep learning architecture using fusion will also be a part of the project scope. The fusion method is a combination of the semantic representation of the different text elements within a shared feature space. This aspect of the project scope illuminates the enhancement of the representation learning and the potential ability of the model to determine numerous applicable labels to a particular article. Other transformer architectures other than BERT are not included in the project scope as they require exploration.

The other vital area that must be included in the scope is the use of multi-label classification techniques that use sigmoid activation functions and binary cross-entropy loss functions. The model is developed to independently categorize more than one label, unlike when it is limited to a single category classification. The analysis of the performance is performed through the standard multi-label measures of precision, recall, and F1-score, therefore making sure that the performance of the model is properly analyzed on the scope.

The implication of this project also includes the implementation of a trained model with the use of the Streamlit framework. The area of application is applied and real-time in nature because the user will be allowed to enter the news articles and receive the projected categories through the web interface. However, it is not the scope of consideration of using the model in a cloud platform, the development of a mobile application and integration of the model with live news feeds.

Lastly, the project is limited to research and academic appraisal purposes. Such topics as the classification of multilinguals in real-time, streaming data processing, and more advanced methods of explain ability are beyond the scope of this project. Nevertheless, the present project provides an excellent start in the further extension and an effective demonstration of a multi-label news classification based on the fusion with transformer models.

1.5 Significance of the Study

The primary objective of this project is to design and build an automated multi-label news article classification system, which has the capability of effectively classifying a news article into a number of thematic labels. As compared to the traditional single-label classification systems, the proposed one will be capable of categorizing a news article into different thematic labels, which would be more realistic and feasible in the current news environment where news articles can represent different fields including politics, economics, business, and international affairs.

The other primary objective of this project is to consider the Reuters- 21578 dataset as benchmark data set to compare the performance of multi-label classification systems. The dataset provided by Reuters-21578, is realistic and generally acceptable as an experimental environment because of its multi-topic labels and the diversity of news articles. The project will focus on a pre-processing phase of the dataset that will provide the efficient pre-processing of the dataset so that the dataset can be introduced to the deep learning classification but the information of news articles will not be distorted.

Among the key objectives is to recognize and deal with the lengthy components of the news articles, i.e., the headline, summary, and body, instead of having a single text piece in the form of the entire article. This way, the system will be in a position to extract section-level semantics, where the headlines will be utilized to summarize the intent of the semantics, summaries will be characterized to highlight the important points, and the body will be utilized to explain the semantics in detail.

The other objective of this project is to play around with a fusion architecture with the help of BERT to extract individually deep contextual embeddings of every piece of the text and incorporate them with the help of a fusion layer. In this manner, the system will be in a position to learn additional semantic representations in the system by determining the relations between various parts of a news article. This objective will directly assist in increasing the accuracy of the classification task.

2. LITERATURE REVIEW

The rising popularity of online news media has created the growing demand of automated methods that will be able to classify a tremendous amount of text-based news data. Therefore, classification of news articles has become a trendy area of research within Natural Language Processing. During the early life research, the focus was on the development of automated methods of sorting news articles into a set of pre-defined categories to create information retrieval, document indexing and content recommendation system. Though these initial steps of the research demonstrated the feasibility of the automated classification, they were not quite efficient when it comes to handling complex and large-scale news data.

Among the most significant findings based on the available literature, it is necessary to note that in the majority of cases, the oldest news classification systems were inclined to think that every single piece of news could be associated with a single category. The truth, however, is the news items usually cover more than one issue simultaneously, including political affairs that are likely to affect the economic situation or technological aspects that are likely to affect business markets.

The traditional machine learning algorithms were more salient in the early phases of the research where manually designed features such as Bag-of-Words, n-grams and TF-IDF were used to encode text numerically. Even though these techniques were computationally effective and could be easily applied, they did not have the capacity to learn the context and semantic association among words. There were also research studies that demonstrated reduced performance when the techniques were applied to large and complex news articles resulting in lower classification accuracy and bad generalization.

Nonetheless, as the power of computers increased, and neural networks were implemented in the form of deep learning libraries, scientists began to use neural network-based models, including Convolutional Neural Networks and Recurrent Neural Networks, to perform the process of news classification. These models turned out to be more efficient in terms of learning tasks in a hierarchical and sequential form based on text. However, it can be seen that the existing literature has shown these models to have problems with long-term dependencies and that they were also not particularly useful in the multi-label classification task (particularly when dealing with large datasets like Reuters-21578).

The introduction of models based on transformers has been a major step in the text classification line. [1],[10]. Transformer-based models such as BERT are based on self-attention mechanisms that allow them to take into account global context and have achieved state-of-the-art performance on multi-label classification tasks [2], [4], [9]. Nonetheless, when news articles are referred to in the present research, they are regarded as a sequence of a text as a whole without looking at the structural aspects like headlines, summaries, and bodies. Recent research shows that fusion-based and multi-stream models which are the integration of multiple representations have demonstrated better performance in the multilabel text classification [3], [5]. Nonetheless, the study of BERT-based fusion models of multi-label news classification on conventional datasets like the Reuters-21578 has not been studied, and this is the focus of the proposed research.

2.1 Overview of Existing Research

The latest increase in the use of digital news media platforms and online news media sites has led to an unprecedented increase in the volume of text-based information in the internet. This increase has created a high need in development of automated systems that can effectively categorize and sort news articles. Accordingly, news article classification has emerged as a significant field of Natural Language Processing research. Most news classification systems created in the early phases of research were created assuming that each news article belongs to one category only. With this assumption, classification became easier and researchers could apply conventional supervised learning to classification.

Early news classification research saw the use of conventional machine learning techniques. The popular algorithms employed in classifying news in terms of naivety, Support Vector Machines, k-Nearest Neighbors and Decision Trees were widely used because they are simple and efficient. These algorithms engaged feature engineering by methods like Bag-of-Words, n-grams and Term Frequency-Inverse Document Frequency. Though these techniques demonstrated average success when used in conventional news classification corpora such as Reuters-21578, certain studies found these techniques to have challenges with regard to generalization in the event they were used on a large collection of news.

The primary limitations that have been found in the current research work are that the traditional feature extraction methods cannot extract semantic and contextual relationships between texts. The Bag-of-Words and TF-IDF models cannot extract contextual significance among texts because they perceive texts as randomized collections of words where the sequence, syntax, and the context of words are not recognized. Studies indicate that this disadvantage is more pronounced when it is used on longer and more complicated news items, on which contextual interpretation is pertinent to achieve proper classification.

As the machine learning algorithms got better, scientists began applying supervised learning models to news classification problems [6],[13]. Naive Bayes, Support Vector Machines and Decision Trees were used as models and Reuters-21578 were taken as benchmarks. These papers demonstrated that machine learning classifiers could be superior to rule-based classifiers regarding accuracy and efficiency. These models were however largely single label models that were not a reflection of real life issues where articles can be attributed to two or more categories.

The second weakness that is present in the literature is that shallow text features are used. Bag of Words, n-gram and TF-IDF were common methods of converting text data into numerical vectors. Even though the methods were computationally effective; they did not consider the rank of words, syntax, and semantics. In some literature it was discovered that models generated with these features have difficulties in finding semantic relationships particularly when it is dealing with large news articles.

Applying the deep learning models was a real advancement in the news classification research. Convolutional Neural Networks and Recurrent Neural Networks were taken into consideration due to the possibility of their automatic hierarchical and sequential learning of text data. Based on the existing literature, it is visible that these models have been in a position to improve the performance of the classification tasks relative to the traditional machine learning methods. However, they were also being constrained by high computational complexity and lacked ability to capture long term dependency especially in large datasets like Reuters-21578.

2.2 Theoretical Frameworks

The history behind the theoretical background of news article classification can be tracked down to theories of text classification and information retrieval which are concerned with the automatic classifying of text documents according to predetermined categories. When applied to news articles, it is in the context of classification: a way of assigning structured text data to significant topic labels using semantic content. Theoretical methods of text classification are as old as the history of studying the field, and text classification was viewed as the problem of pattern recognition, with the statistical features representing a document and the discriminative or probabilistic learning algorithms classifying them.

The other notable theoretical concept that can be applied in this study is the theory of multi-label classification. It is a generalization of the single-label classification problem, in which an example may simultaneously have many labels. Multi-label learning aims at modeling each document as a binary output vector with each output of the document being the presence or absence of a specific label. The theoretical literature that is currently available has demonstrated the fact that multi-label classification is more challenging than single-label classification, because of label dependency, label imbalance, and threshold optimization.

The theoretical basis of the text classification systems has also included the Vector Space Model. The Vector Space Model conceptualizes documents as vectors in a high-dimensional space with each dimension being represented by a term or a feature. The Vector Space Model is the origin of the Bag-of-Words algorithm and the TF-IDF algorithm. The Vector Space Model, though helpful in the task of early text classification, is founded on the independence of words and fails to consider the context semantics, so it is not as effective with complex news articles.

As neural networks are developed, the representation learning theory has played an important role in text classification research. Representation learning is a problem that is interested in learning meaningful features of data without feature engineering. Convolutional Neural Networks and Recurrent Neural Networks are neural network-based models, which are founded on feature learning and modeling in a hierarchy. The models have assisted in acquiring syntactic and semantic patterns in text data which enhanced the functionality of text classification problems in comparison with conventional feature models.

The evolution of attention mechanisms was an important theoretical advance in the sphere of Natural Language Processing. The attention theory postulates that not every word within a document bears equal weight as far as the meaning of the document is concerned and that a model must be capable of attending to parts of a document which are informative. It is a key principle of transformer models, and in this case, self-attention mechanisms can compute contextual representations by modeling self-relationships between all words in a document.

2.3 State-of-the-Art Techniques

The introduction of deep learning and transformer models has spearheaded the recent changes in the classification of news articles. The existing state-of-the-art approaches are mostly centered on acquiring contextual representations of text that do not only rely on the word frequencies at the surface. These models can do much more than the traditional machine learning method in processes like multi-label classification of news articles.

The development of pre-trained language models, such as BERT, RoBERTa, and XL-Net, has become an important breakthrough in text classification. Such models are trained on a large volume of data and could be subsequently fine-tuned to a downstream task. The experiments carried out on the subject have demonstrated that BERT models with fine-tuning have succeeded in providing dependable contextual embeddings that have been useful to improve the accuracy, efficacy, and generalization skills in news classification activities, such as the Reuters-21578 dataset.

Also, the state-of-the-art solutions in the modern world target multi-label learning frameworks. Unlike the case of single-label classification model where softmax activation is used, current solutions use sigmoid activation functions to independently predict a number of labels on a given document. The framework, in addition to binary cross-entropy loss, can cooperate with overlapping labels and class imbalance, which are the common characteristics of real-life news data.

The other important advancement is the use of document structure in classification architectures [7],[8,],[14]. Instead of viewing a news story as one piece of writing, current scholarship has explored the use of structure-conscious approaches that conceptualize headlines, summaries and bodies as distinct entities. Fusion-based architectures take advantage of the section-level representations, and then the model can collectively explore the various semantic facets towards better multi-label prediction.

Lastly, the state-of-the-art models currently focus on deployment and evaluation. Multi-label assessment measures such as precision, recall, and F1-score are being increasingly employed by researchers to assess a model. In addition, that lightweight web frameworks, such as Streamlit, can be used to deploy trained models demonstrates the generalizability of the models and assists in the creation of scalable news classification systems.

2.4 Research Gap Identification

The current literature in the field of news article classification has made significant progress in the use of machine learning and deep learning methods. However, the most important disadvantage identified in the current literature is that most of the studies are focused on single-label classification tasks, where all news articles are assigned to one category. In real life scenarios, the news that is reported is more likely to report on more than one thing at a given time, we have political news which may affect the economic situations, there is also the technological news which may influence business situations. Although there were some studies that tried to research the multi-label classification task, using most of them is founded on a conventional method or shallow models that disregard the complex relationships of semantics in news articles.

The other major gap in research is the fact that the existing models assume that the news articles are a piece of text without structures whereas the natural structures of the news articles like the headline, summary and the body text are not taken into account. These elements have been seen in the literature to be of different semantic significance with headlines giving a concise statement of intent, summaries a statement of key information and the body text a statement of detailed information. This structure has not been exploited by current models.

In addition, transformer models such as BERT have demonstrated superior results in text classification, but there is minimal research work on fusion models of BERT in multi-label news classification with the use of conventional datasets such as Reuters-21578. The majority of the current research literature has employed BERT with single text input or has been single-label classification. An integrated framework to represent the headlines, summaries, and bodies using a fusion layer and evaluate its functionality on the basis of appropriate multi-label measures is required. This study paper fills this gap by introducing a BERT based fusion model to classify news articles in multi-label.

3 METHODOLOGY

The methodology chapter gives a general perspective of the whole process, the methodologies and tools used in the development of the multi-label classification system on news articles. It also gives a summary of the process through which the system will turn raw textual data into useful formats, the integration of textual data in the headline and summary, as well as the textual data in the body of the article, the use of BERT in the creation of the multi-label classifier, and lastly, the deployment of the model.

3.1 Project Workflow

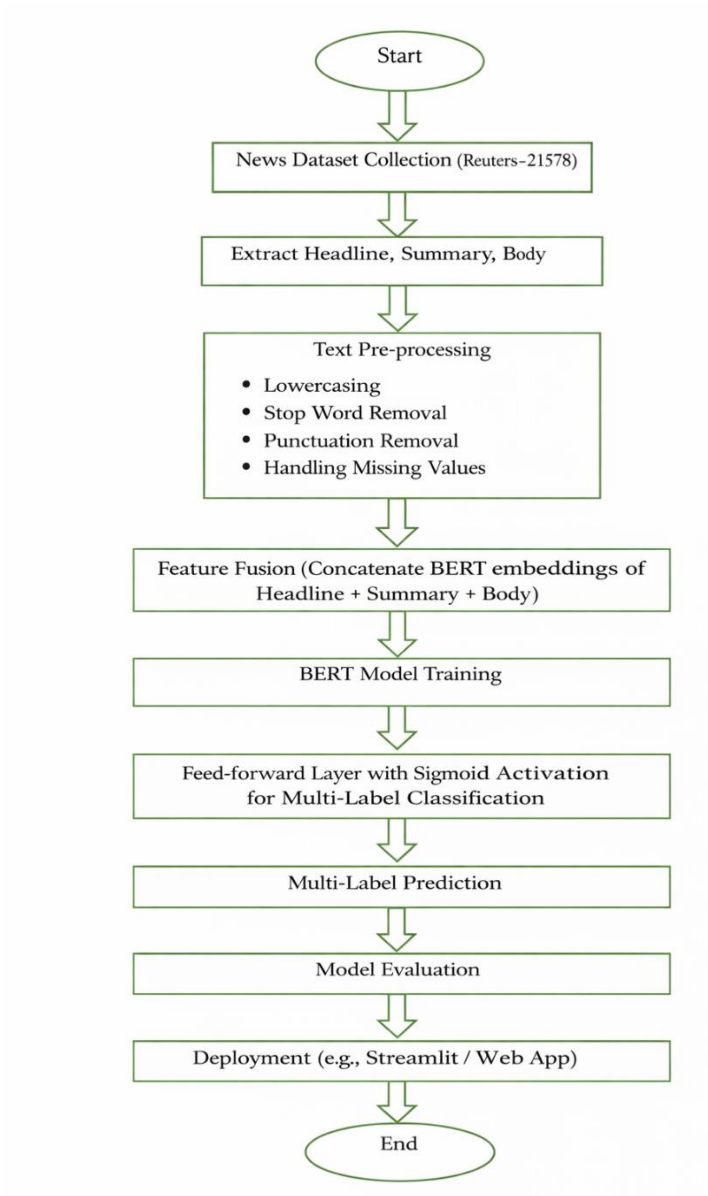


Figure 3.1 Project workflow

The entire workflow will consist of the following steps:

- Dataset Collection
- Data Pre-processing
- Text Segmentation
- BERT- Feature Extraction.
- Feature Fusion
- Multi-Label Classification
- Model Evaluation
- Model Deployment

These stages are explained as discussed below.

3.1.1 Data Collection

The initial project stage is the gathering of the data set of the relevant quality to classify it using multi-label classification. To this end, the data set of Reuters-21578 has been employed. It includes 21,578 news items gathered up in the year 1987 by Reuters newswire. The news articles consist of four significant sections: the headline that comprises the summary of the news article; the summary that has the intermediate contents of the news article; the body, the full text of the news article; and the category, which contains the subjects in the news article. It is believed to be the standard of conducting a text classification research due to the unique nature of the set of data and multi-label news articles integration. As the news articles are already separated into headline, summary, and body, the necessity to separate them manually is eliminated, which will save a lot of time.

3.1.2 Data Pre-processing

Pre-processing is a valuable step as it ensures the textual data is clean; is standardized and is ready adequately. Direct use of textual data contains numerous types of noise in the state of irrelevant characters, figures, stopwords and others, which can affect the model negatively. When the project is launched, the text data is changed to lowercase to ensure that there is uniformity and no differences arise in words. Unnecessary characters, punctuations and numbers are removed since they are irrelevant. In some case some words can be problematic as they are not that meaningful; such words are called stopwords; they are such words as the, is, and, etc.

The application of these stopwords can have an adverse effect on the process of model learning. Another process is called Lemmatization in which the vocabulary of words is narrowed down to base words. In that process, words that convey the same meaning like the word ran and the word running are transformed into their base words. Lastly, the data is tokenized to the form necessary to feed to the BERT model.

3.1.3 Text Segmentation

After pre-processing the text has been done each text of the news is divided into three sections namely headline, summary and text body. To conclude, the headline of the text may be regarded as a reflection of the central subject of the text, providing a general idea of what the text is all about in an effective and fast way. On the same note, summary of a text also adds extra intermediate information which ties the whole text to the underlying knowledge of the headline of the text, giving a medium knowledge of what a news text is all about. Thus, the text segmenting strategy taken by this model enables it to obtain a hierarchical interpretation of the text that is vital in a multi-label classification task since relationships in a text segment and classes are quite complicated.

3.1.4 BERT-Based Feature Extraction.

To get useful semantic features of the various sections of the text, the Bidirectional Encoder Representations of Transformers (BERT) methodology has been employed. BERT is based on the principle of contextual word embeddings, which means that it has the ability to get the meaning of the words depending on the context of the words the word of interest is in. With the BERT approach, the tokens of the various parts of the text are translated to word embeddings with each token in the parts of the text having a different embedding.

The conversion of the tokens to vectors has been done using a combination of token, segment and position embeddings. The BERT approach used will make the word parts of the text represented in the feature space based on the word embeddings to be correctly classified by the classifier since it has a higher capability to produce the context of the words in the text segments in consideration.

3.1.5 Multi-Label Classification

The classification stage is aimed at making predictions based on one or several categories in relation to a particular news article. The output of the combined embedding layer is then taken through a fully connected layer which then undergoes a sigmoid activation function which enables it to make individual category predictions separately. A binary cross-entropy loss function can be used to train the model and will be used to calculate errors that befall each category, thereby aiding in the better comprehension of the multi-label learning method. It is also through this type of multi-label classification that the system is able to understand the intricate relationship that comes with news articles.

3.1.6 Model Evaluation

Multi-label classification should be judged using special metrics. Correctness and completeness in the label predictions of the model are measured in this work by the use of precision, recall, and F1-score. Hamming loss has been determined to quantify the fraction of labels which are misses, whereas subset accuracy quantifies the fraction of samples on which all labels are hits. This performance measure will make sure to make certain that extensive assessment will be accomplished to the model on both the per-label accuracy as well as the overall multi-label performance, which is needed to comprehend the applicability of the model in the real world.

3.1.7 Model Deployment

The last phase of the project will be the implementation of the model into practice. To this end, the library of the Streamlit web applications is employed to produce a web application whose interface provides the ability to add the new news items and classify them. The word-embedding layer transforms the input text into vectors and the pre-trained model is loaded in the backend to pre-process the text and convert it into vectors. These vectors are further combined and sent out through the classification layer to get the category of the news article. The acquired category of the news article will then be presented to the user with the probability of the above category above a given threshold. The model can then be employed by the masses to categorize the news articles so that it can be practiced.

3.2 Data Collection & Sources

The most important aspect of any machine learning-based project is data, the quality, diversity, and accuracy of which are some of the factors to consider in terms of its performance and ability to generalize. In the project at hand, where the researcher has to perform a sorting of the news articles on various categories according to the attributes like headings, synoptic, and bodies, the main concern has to be to search a dataset that is not only all-inclusive but properly organized and precise. Upon the analysis of several various pieces of news-based data, the Reuters-21578 was identified as a source of data that should serve the current project. In fact, the Reuters-21578 is an agreed set used by other scholars and of great relevance in performing text classification and information retrieval tasks, especially because it holds a diversity of news data, which covers various spheres.

The Reuters-21578 data applied in the present case was initially produced by gathering newswire stories throughout the Reuters database during the year 1987. It contains 21,578 articles, and each of them may fall under one or a number of various classifications, such as a business related article may also be classified under the umbrella of politics. This renders the dataset of significant use in the purpose of the project, as the discovery of shared data will be vital in classification. Along with this and the fact that the article is divided into head, summary and body also provide good training data to the model which enables it to comprehend the various levels of text used.

The fact that the content in the Reuters-21578 data has been formatted in such a way that it does not need much manual segmentation or labeling is one of the most critical benefits of utilizing the dataset. Each single article of the dataset includes metadata such as document ID, topic labels and textual content, which is why it is considerably easy to get what is required to be processed with relative ease. The headline is a brief depiction of the article which is good in capturing the high-level context. The summary is an intermediate level textual form of writing that acts as a transition between the headline and the entire body, with the body being full of details that are required in fine-grain classification. This enables the model to acquire both general and specific representations; hence, it enhances prediction of a large number of labels being allocated to a single news article correctly.

The data set itself is publicly accessible, and is available in a variety of reliable sources, such as the Natural Language Toolkit (NLTK) library, and UCI Machine Learning Repository. Transparency and standardization are crucial attributes of the academic and industrial world that are ensured by using publicly available data. One can replicate the methods and test the data with the same set of data easily by a researcher or practitioner and this renders the entire project as credible and reliable. Moreover, the data set is widely applicable to the past or existing study and the performance it gives can be evaluated when comparing it to other standards in the area of text classification.

Further, data collection was checked regarding proper pre-processing and validation through the assistance of other external sources of data besides the Reuters dataset. These were academic articles, official records of text processing libraries as well as online directions in dealing with multi-label datasets.

The inclusion of this kind of references implies that these pre-processing steps like cleaning, tokenization and lemmatization are done appropriately so as to preserve the integrity of the textual content. This kind of process reduces noise and errors which would otherwise lead to less accurate models. By curating and augmenting data with the best practices performed by previous researchers, the project is sure to produce high-quality data that will give the BERT-based multi-label classification model a maximum opportunity to achieve success.

Lastly, it would be an ideal selection of a dataset to use in this specific project since it is well suited to implement more complex types of Natural Language Processing, like BERT. BERT is recommended to be used with structured data; in this respect, dividing the news articles into headlines, summaries and body would simplify the process of generating lexical tokens needed to create meaningful word embeddings, thereby enabling the successful execution of multi-class classification tasks on the basis of the rich context embeddings which the approach outlined above produces.

Overall, the selection criteria used by the quality and usability of the Reuters-21578 dataset as the primary source of data is a great starting point of the whole project. Its multilabel data properties, data division, domain focus, open accessibility, and its use as a research source data previously all guarantee the choice of this particular dataset as the ideal candidate to be utilized in creating a fantastic and precise system of multilabel news article classification. This makes sure that the whole BERT model used can access any good text based information to make predictions more effective.

3.3 Data Pre-processing

The pre-processing of data is crucial to the Reuters-21578 dataset, as it will be ready to classify news articles into hierarchical and multi-labels with the help of a BERT model. Although it is structured, it may have a lot of noise, inconsistencies, or variations in terms of form, which may be debilitating to the performance of the model. The primary objective attained during the process of data pre-processing is to clean, normalize, and organize the unstructured textual data in a way that enables the BERT model to extract meaningful output or more to the point meaningful embeddings upon which further classification can be performed. The pre-processing part of data consists of individual portions of news, such as headlines, summaries, and body in a hierarchical format so that context is maintained on all levels in a news article itself.

3.3.1 Text Cleaning

Removal of irrelevant contents and noises in the text is the first step in pre-processing. The news article can include punctuations, special characters, numeric strings and HTML artifacts which may not have semantic meaning.

- **Elimination of Punctuation and Symbols**: Punctuations and symbols like commas, periods, semicolons, quotation marks, and others are eliminated like @, # and &.
- **Numeric Removal**: Removal of numeric values which do not make sense in the context is done to ensure that the model does not memorize irrelevant patterns.
- **HTML/Markup Cleaning**: Any HTML/markup traces left behind by the dataset are removed to just leave the text.

Text cleaning makes sure that only meaningful words are left and this makes the embedding quality better and the noise in the model input is minimized. The articles are separated into each section (headline, summary, body), and cleaned to retain the structure of the data.

3.3.2 Text Normalization

The text normalization process subsequent to the cleaning process standardizes the text by decreasing the level of variation in.

Upper case to lower case: All words are reduced to lower case letters. It is done to provide uniformity since it considers terms such as Economy and economy as the same.

- **Whitespace Handling:** It eliminates extra space or tab or new line character to be formatted appropriately.
- **Expansion Contraction:** This is a rule that expands on shortened words, e.g. "don't" is changed to "do not," "isn't" is changed to "is not" etc., enhancing semantic unity.

Normalization helps the model to focus on meaningful patterns and not the superficial differences in text format.

3.3.3 Stopword Removal

- **Stopword:** Words that are common such as: the, is, and, and, among other common words, but carry less semantic significance. They are eliminated in order to make the model more efficient.
- **Selective Stopword Removal:** The default set of stop words are eliminated, although domain specific stop words, such as market, stock, or government are retained, since they have very big discontinuous impacts on news classification.
- **Effects:** brings attention to content words and this enhances the discriminative power of the BERT word embeddings.

All the steps of the stop words elimination are performed on headlines, summaries, and body parts.

3.3.4 Lemmatization

The process of lemmatization simplifies the words to their root so as to eliminate redundancy and create semantically consistent words.

An example is that, "running," ran and runs are all transformed to run.

- **Merits:** The dimensionality reduction is a good thing as opposing representations of the same word will be linked to the same embedding.
- **Application:** It is applied to all the segments to give uniformity in the information contained in the headline, summary, and body.

The step is critical in BERT since subword tokenization is normally improved with word forms that are normalized.

3.3.5 Tokenization

The process of representing normal text as a series of BERT subword tokens generated is called tokenization.

- **BERT Tokenizer**: It is based on the idea of dividing the text into subwords.
- **Special Tokens:** Initially, CLS token is added to indicate the sequence and the SEP token is added to the end as a segregating segment.
- **Padding & Attention Masks:** Padding is done to ensure that the sequences are as long as possible and then the attention mask is used to prevent padding during embedding computation.

This will guarantee that BERT is capable of processing hierarchical information in all the three parts effectively.

3.3.6 Tackling Multi-Label Representation.

Since it is multi-label data, the categories are coded in binary coding, where a dimension is one category:

- 1 indicates that there is the category.
- 0 means that there is no category.

As an example, the article can be classified under finance and politics, and therefore the vector would have a value of [1, 1, 0, 0, etc...]. Particular attention is paid to the imbalance in classes, which can be underrepresented in the issue. Effective learning of underrepresented classes in the datasets is determined by the use of weighted loss functions.

3.3.8 Segmentation and Sequence Length Optimization.

There are changes in the body, summaries and headlines in the sense that they differ in length and this makes embedding extraction difficult.

- **Maximum Sequence Lengths**: Each segment is defined separately to be able to control memory usage.
- **Padding:** This refers to a situation when shorter sequences are brought up to equal length.
- **Truncation:** Longer sequences are truncated with much care such that the content of significance is not lost.

- **Attention Masks:** It is important to ensure that token padding is not given to the embedding calculation.

The alignment provides uniformity in the tensors that come in and feature fusion hierarchical relationships.

3.3.9 Summary of Pre-processing

In short, we can conclude that we have a pre-processing pipeline that we apply to pre-process our raw Reuters-21578 newspaper articles, and transform them into a clean, normalized, and semantically-rich input with which we run BERT-based multi-label classification.

- Text Cleaning (removal of punctuations, numbers, html tags, etc.)
- Normalization of the texts (Lower Case, Space Handling)
- Eliminate stop words (domain sensitive)
- The process of splitting a word by its root into parts is called Lemmatization (Root Word Conversion).
- Bert Model sub-word tokenization, special tokens, padding and attention masks.
- Multi-Label Representation With Binary Vectors.
- Alignment of Segments and Length of Sequence.

This, consequently, would make the embedding quality to be of high quality with no noise of any kind added to it and the semantic context maintained. This would lay the foundation of any feature extraction or hierarchical classification that would be necessary in achieving maximum accuracy in the classification of news articles.

3.4 Proposed Model

The proposed model for the classification of news articles into multiple labels optimizes the use of BERT-based embeddings as well as a hierarchical feature fusion method to classify the news article based on the headlines, summaries, and bodies of the text. The model is capable of capturing the semantic relationship of the text across different segments of the text to classify the same into multiple labels. The model is optimized with pre-trained BERT embeddings to utilize the advantages of the pre-trained model.

The process starts with data pre-processing, which involves removing noise from the dataset, tokenizing the text, and then converting it into numerical embeddings that can be used as input for deep learning models. The pre-processed data is then used as input for a fine-tuned transformer model that has been trained using supervised learning with a multi-label encoding strategy.

The output layer uses a sigmoid activation function to predict independent probabilities for each label, which enables the system to predict multiple labels at once. The model is trained using binary cross-entropy loss to ensure that the classification is accurate for all labels while preventing overfitting using regularization and validation.

To ensure that the system is user-friendly and accessible, the trained model is used in a Streamlit web application that enables users to input their own news text and receive instant predictions for categories.

The application combines the use of the tokenizer, the trained model, and the label encoder to provide a seamless experience. This application of deep learning has the ability to be scaled and used in real-time, making it a feasible tool for content organization, media monitoring, and intelligent information retrieval.

3.4.1 Model Architecture Overview

There are three significant stages of the architecture:

- **Embedding Extraction, Segment-wise:** Each of the following segments of the article, including the headline, summary, and the body are fed into the already trained BERT model and yield rich contextual representations.
- **Feature Fusion Layer:** This layer entails a concatenation of the embeddings of all the three segments through the attention-based fusion mechanism to obtain a single feature vector that represents the whole article.
- **Multi-Label Classification Layer:** The latent multi-featured data is passed through a single (or multiple) fully connected layer, which is then passed to a Sigmoid activation function to decide whether one or more of the categories are present or absent.

The same architecture further makes sure that the model relies on both the segment-specific information and global context information to classify data, hence the output of multi-label is more precise.

3.4.2 Embedding Extraction using BERTs.

The central part of the model relies on the Bidirectional Encoder Representations of Transformers to produce embeddings of every piece of an article. BERT embeddings have some important characteristics that are:

- **Bidirectionality**: BERT draws context of both the left and right sides of any token to be able to represent subtle nuances of meaning.
- **Pre-trained Knowledge**: BERT is also pre-trained with large corpora such as Books Corpus and Wikipedia that have enabled it to understand grammar, semantics and syntax.
- **Segment Specific Embeddings**: Different embeddings are generated on headline, summary and body so that there can be hierarchy context.

All the sample segments are tokenized by a BERT tokenizer, converted to input IDs and attention masks and then run to the BERT model. There are many other approaches, but most frequently one will take the [CLS] token embedding of every segment as the representation since it was proposed to represent the semantic meaning of a complete sequence.

3.4.3 Feature Fusion Mechanism

After capturing the embedding of each individual segment, there is a process known as a feature fusion mechanism, which fuses all the three vectors to a single representation. The fusion may be done in different ways:

- **Concatenation:** Concatenating the [CLS] embeddings of headline, summary and body.
- **Weighted Fusion:** The segments have the weights assigned in accordance with their relevance in classification.
- **Attention-Based Fusion:** This aims to learn the score of attention respectively on every segment to emphasize the informative sections.

The combination of the meaning of the entire article provided by the resulting vector tackles not only the general concepts reflected in the headlines, but also the minor details reflected in the text of the body. This hierarchical combination is especially important in order to use it in multi-label classification, when certain terms may be prevalent in a certain section of the text (e.g., the word "finance" may feature in the body, whereas the word politics appears in the headline).

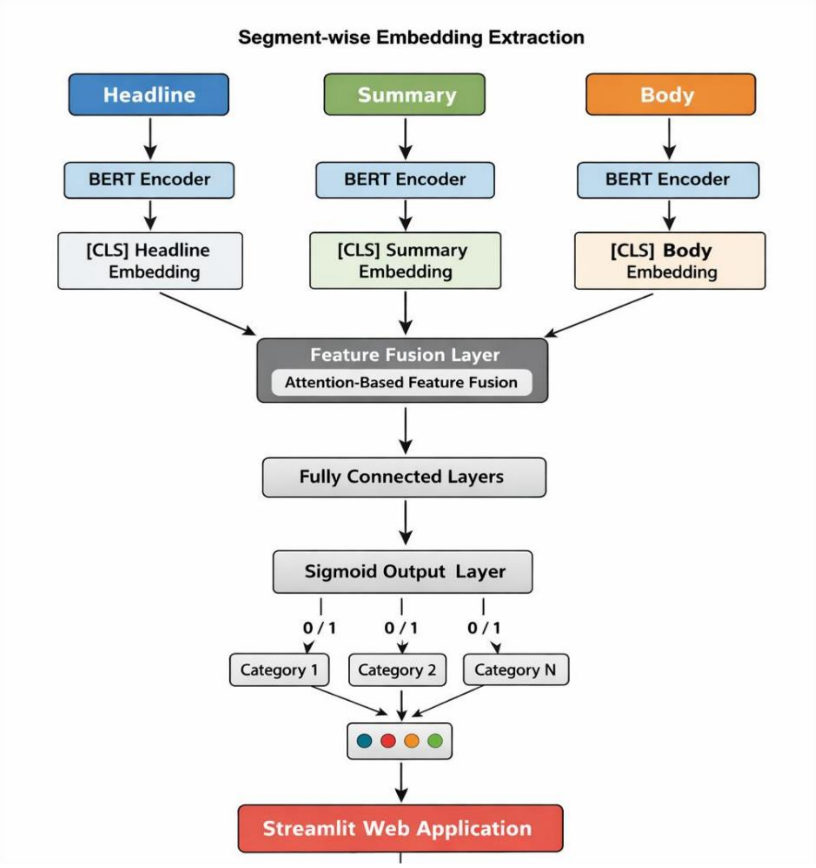


Figure 3.4 Flowchart of architecture

The diagram 3.4 represents a text classification system with many labels in which three parts of an article (Headline, Summary, and Body) are individually encoded to extract [CLS] embeddings using BERT before fusing these with each other using an attention-based layer of feature fusion.All these fused elements are processed through the fully connected layers and a sigmoid output layer to generate binary predictions between several categories and the results are delivered by a Streamlit web application.

3.4.4 Multi-Label Classification Layer

Multi- label classification layer: Multi-label classification layer is an extension of the previous classification layer. Then, there is a neural network that processes the combined features. One or two dense layers are likely to make this network, with dropout to avoid overfitting. On the last layer we have N neurons and N is the total number of classes in the data set.

Activation Function: The output neuron has a sigmoid activation function to produce a probability between 0 and 1 of each individual class.

Loss Function: BCE Loss function is applied in the comparison of the probabilities and the true labels. This loss is utilized in multi-label data.

The predictions are of a multilabel type where a post may fall under various categories. One can also threshold the predictions (say probability > 0.5).

3.4.5 Model Smoothing and Optimization.

To enhance generalization, several methods have been used:

- **Dropout**: This is an applied method that is found in the classification layers but is a way of reducing overfitting by randomly removing nodes.
- **Learning Rate Scheduling**: AdamW optimizer is applied with a low learning rate of 2e-5 which is effective in fine-tuning the BERT model parameters.
- **Early Stopping**: Early aborting the training can be done by monitoring the validation loss metrics by making sure that no improvement occurs. This helps to counter overfitting.
- **Gradient Clipping**: This is applied to gradient restriction whereby, it helps to avoid huge gradients during a backpropogation.

Under these strategies, the training is stable and the model can generalize to unknown article.

3.4.6 Mathematical Function

Below, the suggested multi-label news article classification scheme will be mathematically developed considering processing an article headline, summary, and body with the help of a BERT structure that uses attention-based feature fusion. A news item can be characterized as follows:

$$A=\{H,S,B\}$$

H represents the headline, S is summary and B is body of the article.

The pre-trained BERT encoder is used to encode each of the segments individually. This operation may be explained mathematically as:

$$EH=BERT(H)$$

$$ES=BERT(S)$$

$$EB=BERT(B)$$

Where EH and ES and EB denote the embedded parts of the context of the [CLS] token in the output of the BERT model with reference to the headline, summary and body respectively. To combine information in all the three segments, an attention-based feature fusion approach is used. Under this approach, weights of attention of all three segments will be calculated to ascertain their weights.

$$aH,aS,aB=Softmax(Wa[EH;ES;EB])$$

And [;] is the concatenation of vectors, and Wa is the attention weight matrix. F is then computed as fused feature vector as follows:

$$F=aHEH+aSES+aBEB$$

3.4.7 *Pseudo Code*

The step wise working of the proposed BERT-based multi-label news article classification system that involves a headline fusion, summary fusion and the fusion of the body is presented in the following pseudo code.

Input: News Article (Headline H, Summary S, Body B)

Output: Predicted Label Set Y

Begin

Loading BERT model and tokenizer pre-trained models.

Pre-process input text:

a. Clean H, S, B

b. Split all the segments with the help of BERT tokenizer.

c. create attention masks and pad.

Extract embeddings:

a. Pass H through BERT - obtain EH

b. Pass S through BERT - obtain ES

c. B by BERT - receive EB.

Perform feature fusion:

a. Concatenate EH, ES, and EB

b. Calculate attention weights of every segment.

c. Form fused feature vector F.

Classification:

a. Pass F across layers which are fully connected.

b. Sigmoid activation to get probability scores.

Label prediction:

a. For each category i:

If probabilityi [?] threshold t

Assign labeli = 1

Else

Assign label_i = 0

Return predicted label set Y

End

3.5 Experimental Setup

The whole research methodology has the experimental setup as the backbone that establishes an appropriate setting of validating and implementing the proposed hierarchical BERT-based multi-label news classification model. This experimental set-up is aimed at ensuring reproducibility, reliability, and transparency of the experiment or study. The experiments were done to determine the ability of the model to effectively learn the segment specific features based on the headlines, summaries and body texts and provide accurate predictions of the different categories of the Reuters-21578 data.

3.5.1 Hardware Environment

Transformer-based models, e.g., BERT, have a high number of parameters and attention mechanisms, leading to the high computation load of the model. The following was the specifications of the system on which experiments were conducted:

- **Processor:** A multi-core processor such as Intel core i7-10700K will be required to execute this application effectively in terms of pre-processing and tokenization.
- **Graphics Processing Unit (GPU):** Graphic card NVIDIA RTX 2080 Ti / 3090 GPU with 11-24 GB VRAM. During the process of fine-tuning, GPUs are significant in accelerating the forward and backward pass of the BERT model.
- **RAM:** 32 GB of RAM or above to load data, compute pre-processing, as well as the calculations of intermediate tensors.
- **Storage:** 1 TB SSD to be used in supporting quick read/write operations on the dataset files, model checkpoints, and inputs.

The hardware setting provides that large series of words of body text can be proficiently broken down and that the hierarchical consolidation of segment embeddings do not break down owing to memory constraints.

3.5.2 Software Environment

The software environment is designed to deal with text processing, model building, training and evaluation and deployment.

Programming Language Python 3.9. Python has been chosen because it is well-suited to other machine learning libraries.

Core Libraries:

- **Transformers (HuggingFace)**: This package provides is a pre-trained model of BERT, tokenizers, and an embedding.
- **PyTorch:** This is a deep learning framework used to define, train and fine-tune the BERT based model.
- **NumPy and Pandas:** Assistance in the effective processing of information, label matrices and models.

- **Scikit-learn:** It is applied in the splitting of the dataset, binarizing the label, and calculating the metrics such as the precision, recall, and F1-score.
- **NLTK:** It provides different tokenization, stopword and lemmatization functions.
- **Streamlit:** It is applied to deploy the trained model in a manner that is reachable by the end user.

With this kind of a combination of libraries, it becomes easy to manage the data pipeline, model it, and deploy it with ease to be tested in real time.

3.5.3 Preparation and Splitting of Data.

The second most critical property of an experimental setup is the corresponding property of having a representative training, validation, and test sets in that the datasets are multi-label. Reuters-21578 dataset has 21 and 578 news articles that are classified under one or multiple categories.

- **Training Set (70%):** This set is used in BERT embedding training and identification of connections between the text features and categories.
- **Validation Set (15%):** Only used during training, but in a controlled fashion, and the primary purpose of it is to avoid over fitting, adjust parameters and enforce early stopping.
- **Testing Set (15%):** Evaluates the end performance of the model designed and provides impartial data.

The stratified sampling preserves the percentage of data in every category, which in the context of multi-label data is highly significant since it is possible that certain categories, such as, finance, are represented significantly as compared to other categories, such as, wheat.

Moreover, there are also no entries of articles without the sections or empty ones because it is guaranteed that only high quality and reliable input is used.

3.5.4 Hyperparameter Configuration:

Hyperparameter configuration is a component of parameter estimation and control, as well as a parameter estimation technique.

To tune the BERT and the Fusion network, the following hyperparameters need to be properly tuned:

- **Pre-trained Model:** bert-base- uncased, containing 12 layers of transformer and 768-dimensional.
- **Batch Size:** This varies between 16-32 determined by the memory capacity of the GPU one is training the model with; smaller batch size use less memory on training but can take more time.
- **Learning Rate:** The optimization value is 2e-5 and the optimizer of choice is AdamW that is often used during the fine-tuning of transformers.
- **Epochs:** 5-10 epochs.
- Early Stopping This happens when validation loss is not improving after 2 consecutive epochs.
- **Dropout Rate:** the fully connected layers have a dropout rate of 0.3, and this was used to avoid overfitting.
- **Sequence Lengths:**
- **Headline:** 30 tokens
- **Summary:** 100 tokens
- **Body:** 512 tokens (max. number of single sequence of BERT)
- **Loss:** Multi-label prediction Binary Cross-Entropy (BCE).
- **Cut off point to make predictions**: 0.5 probability. This may be specified based on the precision recall curve trade-off.

The above hyperparameters were computed after conducting an empirical experiment aimed at calculating fast convergence, high accuracy, and generalization of the hyperparameters.

3.5.5 Tokenization and Pre processing

Pre-processing is the very important aspect of the experimental setting.

- **Text Cleaning**: Clean up punctuations, numbers, HTML objects and additional whitespace.
- **Normalization:** White space, contraction expansion and lower casing.
- **Stopword Removal**: Domestic words are removed and domain specific ones that are essential in classification are retained.

- **Lemmatization:** This method involves the removal of redundancies through transformation of words to their root.
- **Tokenization:** Tokenization refers to the breaking up of words with the word tokenizer of BERT into subword tokens, the insertion of the [CLS] and [SEP] tokens, and the creation of attention masks.

The length of sequences in each segment is handled with great care, trimming and filling them out to ensure that the tensors used to feed BERT are equal in length.

3.5.6 Training Procedure

This is done in a systematic and hierarchical order:

- The processed headlines, summaries and body text are broken down into token IDs and attention masks.
- The individual pieces of the text are inputted directly to the pre-trained BERT model to retrieve [CLS] embeddings.
- The segment embeddings are concatenated with attention based layer.
- Multi-label classification is then done through these features being passed through fully connected layers and dropout.
- The calculated BCE loss is in comparison with the actual labels, and the weights are optimized using Adam optimizer.
- Adjustment of the hyperparameters and early stopping is done using the validation loss. The training is kept going until convergence or other termination criteria are achieved.

3.5.7 Evaluation Metrics

The model performance is normally measured against the following multi-label measures:

- **Accuracy:** The accuracy of the predictions of the positive labels.
- **Memo:** The actual number of positive cases.
- **F1-Score:** Precisions and Recalls harmonic mean.
- **Precision:** The percentage of correctly labelled data.
- **Hamming Loss:** Percentage of wrongly identified samples of samples.

Measures are computed on a category and general average basis which provides information of the behaviour of the model on frequent and infrequent categories.

3.5.8 Streamlit Deployment testing

The trained model is pushed and run using the Streamlit framework to test and deploy the model in the real world to guarantee its usability. The values of the model weights are inputted to the Streamlit application. In the Streamlit app, the user will be able to enter a news article headline, summary, and article body using text boxes. This text is tokenized and follows the same procedure as the initial training data to generate a multi-label prediction with the trained BERT based model. The interface is also rendered dynamically with the predicted categories. By doing so we are testing the deployment of the trained model and whether it will be usable in the real world when applied to the actual problems.

It is accompanied by the deployment of the trained model on the real-time setting to test it:

- The model weights that have been saved are loaded into Streamlit.
- The user is able to type in the headline, the description and the content of the news.
- Pre-processing, embedding extraction, feature fusion and classification are carried out in real-time.
- The results of multi-label prediction are shown dynamically on the Streamlit interface, which attests to the real-life applicability of the model.

This will make the model sound theoretically but will also be prepared to application.

4. IMPLEMENTATION

The present chapter realizes in detail the Multi-Label News Article Classification system proposed based on BERT with Headline, Summary, and Body Fusion. The last chapter outlined the theoretical approach to methodology as to how the approach and system architecture were to be implemented; this chapter is the implementation of the ideas in software programming, model building, training process, and mechanism of deployment. The implementation phase is significant in terms of feasibility and effectiveness in testing the suggested approach with the decision to move its design into the working system.

The primary goal of this chapter is to unveil how the BERT-based architecture has been developed and integrated with the intention of making multi-label classification to the Reuters-21578 dataset. The segment-wise embedding extraction has also required the incorporation, which has necessitated the attention-based feature fusion in turn the multi-label classification mechanism. Elaborating every step of the programming process has been a must, so that a person gets to appreciate the entire process of transforming raw data into an useful one. Furthermore, it has been made sure that the overall process can be scaled in order to be updated in the future.

Moreover, the challenges of implementation that are experienced are also indicated in this chapter such as the use of long textual sequences, resource constraints as well as the consistency of the training and the deployment environment. The solutions adopted to address the implementation issues such as some of them are discussed to show how reliable the system would be. This will ensure that the model implemented is running well without compromising the required levels of accuracy.

Also, the chapter discusses the deployment of the learned model into a usable web interface with the help of Streamlit, a framework that is specifically designed to do this. The deployment part allows the user to engage with the site by providing input fields with headlines, summaries and body content and showing real-time predicted news categories of the input content. This deployment property basically shows the practicability of the developed system not only in intra-organizational use but also in the real world to include applications like the possible news recommendation engine. On the whole, one can state that Chapter 4 serves as the bridge between the theoretical base and the experimental assessment of the suggested system.

Based on the description of the implementation procedures, tools and mechanisms, this chapter provides the whole picture regarding the implementation processes and mechanisms of the proposed multi-label classification model. The further paragraphs of the very chapter are dedicated to the explanation of the individual implementation procedures and tools, which makes the general picture of the implementation aligned.

4.1 Model Development

The development of the model is completely aimed at the process of the implementation of the suggested BERT-based multi-label news article classification model in the form of a structured and systematic manner. During this stage, the methodology-based conceptual design, discussed above, is translated into a practical deep learning model that can in fact work with the actual data of the real world. The process of development is believed to be conducted in a way that it is, a modular process i.e. the components like the preprocessing, embedding extraction, fusion, and classification etc., can be created separately.The pre-trained language model is selected as per the research question and the nature of the study.

4.1.1 Pre-trained Language Model

The selection of a suitable pre-trained language model is the first step in developing a model. The BERT-base-uncased model will be used since it has shown very high performance in different tasks on natural language processing. Bert is a deep bidirectional transformer model which acknowledges relationship of contextual links among words by taking both forward and backward flow of text into account. It enables the model to gain more semantic information of each word, which is very important in news articles, as most of the news articles have more complicated sentence structures and semantic dependencies. A pre-trained version of BERT also lowers training time and computation cost by a significant amount, and carries an enriched contextual embedding that increases the accuracy of classification.

4.1.2 Representation and Segmentation of input.

The Reuters-21578 data is automatically divided into three parts of a news article:headline, summary, and body. This is inspired by the fact that various sections of a news article have varying information content, headlines provide an overview, summary the important facts and the rest of the body is more explanatory. The division of these enables the model to maintain semantics of sections.

We feed each of these pieces as an independent input stream, thereby letting the model learn a lot of fine-grained representations that would be lost in a situation where the entire article is fed into the model as a single non-distinguished lump of text.

4.1.3 Tokenization and Input

After segmentation, the text segments are then tokenized with BERT tokenizer. The tokenizer uses the text to extract subword tokens and then converts the numerical tokens in the BERT model into tokens. The tokenized input consists of special tokens, e.g., [CLS] and [SEP]. They are utilized to depict the starting and ending of the sequences. Then the attention masks are developed to distinguish between the important tokens and the padded tokens. After that, padding and truncation are done to make all input segments to be of equal length, such that they can be efficiently processed in batches.

4.1.4 Embedding Extraction by Segments

The coded part of the headline, summaries and body is separately input into the pre-trained BERT. The embedding of the [CLS] token is obtained out of each of the separate outputs of the BERT encoders. The embedding of the [CLS] token holds a summary of the contextual information and semantics in the whole body of input text. After the segments of the headlines, summaries, and body of the news article have been extracted and their individual embeddings have been extracted, it assists the model in differentiating between the categories in the level of the information.

4.1.5 Feature Fusion Mechanism

The word embeddings obtained by extracting the headline, summary and body are combined using an attention-based feature fusion algorithm. This enables this model to give varying weights to different segment derived word embeddings, as they vary in how they contribute to the final classification. This is contrary to other techniques, where the three parts are put together. There might also be more word embeddings in the text (e.g. financial news), or required by need in the text (e.g. breaking news). The integrated feature vector encompasses the general knowledge of the whole article which also incorporates local and global features.

4.1.6 Design of Classification Layer :

This is followed by feed forwarding the combined feature vector through one or more fully connected (dense) layers to do classification. These layers reduce the high-dimensional feature representation to set scores of each category.

In addition, regularization in between the layers of the drops averts overfitting and enhances the generalization and the final output layer has a sigmoid activation that is applied to that layer that provides independent probability values of each category. The design will be relevant in multi-label classification, where an article can be in more than one category simultaneously.

4.1.7 Loss Function and Optimization Strategy

Binary Cross-Entropy (BCE) loss is a function loss that is used to train the model appropriately. This loss serves well in checking the error generated during prediction of one label, and therefore it can be used in a multi-label classification issue. The parameter of the model is to be optimized in the best way, and on this account, the introduction of the AdamW optimizer is presented by its ability to address the sparse gradient and avoid overfitting with the help of weight decay. A suitable learning rate is selected, making the model converge steadily in the process of fine-tuning the complete BERT model and classification head.

4.1.8 Model Training Control and Checkpoint

To mitigate the stability of the model, elaborate mechanisms have been provided to control the modeling process. This will involve the incorporation of a training control mechanism. This process makes the model not to overfit. This is by storing the parameters of the model at periodic steps making it feasible to choose the best performing model in the light of the validation parameters. The validation loss is used to determine that the model is not overfitting.

4.2 Model Training & Tuning

When training and tuning the models in question, a certain emphasis is put on the optimization of the suggested BERT-based architecture to yield a high level of precision and stable results in terms of generalization when it comes to the multi-label news article classification problem.

Here, the training process of the previously processed model is done by giving the input and the performance parameters are refined to achieve excellent performance by the classification to give the output. To adequately categorize the news items, the fine-tuning and training are conducted to get the model stable.

4.2.1 Training Strategy

The training is performed in a supervised learning manner in which the model is trained to map the input text segments to the category labels of those segments. In every training step, head line, summary, and body text is transferred to the BERT encoder to obtain segment-wise embeddings.

These embeddings are combined with the help of the attention-based mechanism and are filtered through the classification layers in order to provide the predictions. Ground truth labels are used to compare the predicted outputs against ground truth labels to obtain the training loss. Then backpropagation is carried out to adjust the model parameters and with time, the model learns meaningful patterns of the data.

4.2.2 Loss Function

As the problem is a multi-label classification problem, it is used to optimize the Binary Cross-Entropy loss. The Binary Cross-Entropy loss considers each of the classes simultaneously and multilabl predictions are possible. Moreover, this loss operation is especially applied to the classes which are not mutually exclusive or independent like news items where the subject is a combination of labels. The loss will be determined per batch and averaged on all classes.

4.2.3 Hyperparameter Tuning

Hyperparameter tuning is another significant component of training. The values of hyperparameters i.e. learning rate, batch size, dropout rate, number of epochs etc. are optimized automatically to arrive at the optimal mixture of values. Better smooth tuning of the BERT model is adopted at a learning rate of 2e-5 without the option of abrupt parameter updating. The batch size is considered based on the GPU with which the models are being trained. The model has a dropout to prevent overfitting. Stopping of training the model is performed through early stopping.

4.2.4 Fine-Tuning the BERT encoder

The BERT encoder will be fine-tuned and the classification layers adjusted in this project to the task of classifying news articles in accordance with the domain knowledge. Fine-tuning will allow the model to modify its word embeddings with references to the language characteristics of the news field. To prevent the occurrence of a phenomenon referred to as overfitting observed to happen at times when fine-tuning is conducted, the learning rate value of the BERT layers will be lower than that of the classification layers.

4.2.5 Performance Monitoring and validation

The train dataset is used in training, the validation dataset is the one that is used to monitor the performance, and the tuning decisions are guided using it. The model is tested on the validation set after every epoch and the metrics it performs are: validation loss, precision, recall, and F1-score. These measures provide the understanding of the learning of the model and overfitting. Early stopping is also adopted using non smoothened validation loss to early stop when it fails to enhance improvement over a predetermined number of epochs, to guarantee good generalization capacity of the model.

4.2.6 Multilabel Prediction Threshold Selection.

In the situation of multi-label classification, the proper threshold of prediction score is highly significant. A default threshold of 0.5 has been given in the above code. This means that when a multi-label classification problem is involved, the labels whose probability is given of 0.5 or above 0.5 are said to be present. Tuning of the threshold of making predictions is carried out to achieve improved precision-recall trade offs.

4.2.7 Stability and regularization of training

A number of regularization methods are applied in order to achieve stable training. This has dropout layers which facilitate over relying on some neurons. Gradient explosion has also been avoided by gradient clipping, particularly when optimizing transformer layers. The AdamW optimizer has weight decay assisting with parameter regularization. All these guarantee the stable and effective deep learning model training.

4.3 System Design

The system design section expounds on the structural design of the proposed multi-label news article classification system. A summary on the interactions between the different sections of the proposed classification system including, input of data, pre-processing, model execution and, lastly, model results are appropriately reflected in this section. The structural formality that is suggested in terms of the designing multi-label news classification system is hierarchical and modular form of the model to allow easy interaction between the backend model and the frontend model interface.

4.3.1 Overall System

Regarding the system architecture, the entire system architecture will be developed as a multi-layered system comprising of an input layer, a processing layer, a model layer and an output layer at the end. Regarding the input layer, the news items will be offered in the form of headlines, summaries, and bodies. Then the text pre-processing and tokenization will be carried out at the processing layer. The BERT-based embedded extractor, the attention-based fusion module and the multi-label classification layers will all be incorporated in the model layer. Lastly, the predicted news categories will be displayed with the aid of the output layer.

4.3.2 Input Module Design

The input module is involved in the acceptance of the news articles text through offline or online sources of the user. The information will be sampled on the Reuters-21578 dataset to provide both training and testing information. The input module will be integrated with a Streamlit user interface to input news article headlines, summaries, and bodies manually, to be deployed. This will also be done under input validation to make sure that no faults are experienced in this aspect.

4.3.3 Pre-processing/ Feature Engineering Module

Pre processing module This process involves the processing of raw data to be consumed by the model. Cleaning operations in this module are the removal of special characters, normalization, removal of stopwords and lemmatization. BERT tokenizer is used to tokenize after all the cleaning operations. BERT is able to do feature engineering automatically and there is no need to extract context embeddings, which makes it avoid feature selection. It is worth mentioning that the entire operations of this module are to make data come in a standard form to enhance the reliability of this model.

4.3.4 Model Layer

A significant part is the model layer of the system. This layer consists of three parallel BERT encoders that have the role of encoding the headline, summary, and body portions of text separately. The contextual information contained in the text embeddings obtained is particular to a block of text. These learned text embeddings are sent to the attention-based fusion layer in order to receive a fused representation of text-wise features. Multi-label prediction is made on the representation obtained here. It is a layer of this model that enables training and prediction without altering the structure of the same.

4.3.5 Fusion of features and Decision Module

The feature fusion and decision module is critical in the merging of the information held in different parts of the text. The significance of each segment is computed dynamically in relation to the output of the attention based weighing technique. The fused feature vector generated is taken through the fully connected layers to achieve the predictions. Each of the nodes in the output layer is operated by the sigmoid activation function to produce independent probability values of all the classes. This module aids in the ability to capture a local and global context with a multi-class classification task.

4.3.6 Backend and Frontend Intergration

The streamlit framework takes care of the integration between the frontend and the backend. The model, which is written in the BERT algorithm, and the logic of model prediction are found in the backend. Conversely, the user interface is simple in the frontend. A combination of the backend and frontend will enable the user to enter the data in the system, and it will be worked on by the backend and the labels of the data will appear in real-time.

4.3.7 System Workflow Design

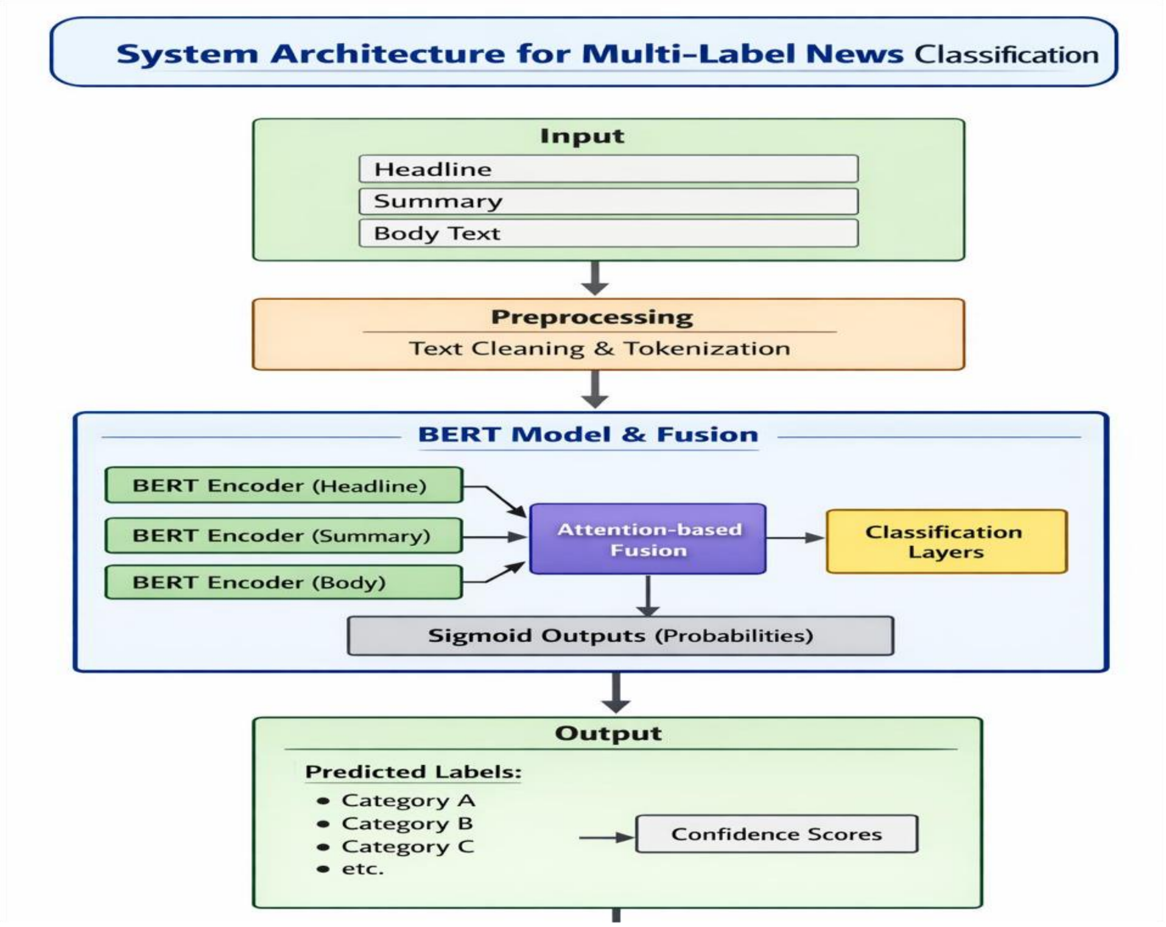
The workflow is divided into the following components: input acquisition, pre-processing and tokenization of the input. In addition, the processed inputs are provided into the model layer to be embedded with extraction, fusion, and classification. The model results are further processed and presented to the user. In such design, the whole workflow is designed to be linear and not containing numerous detours to minimize latency during real-time inference. The workflow design would be clear to make it easily understandable and debuggable.

4.3.8 Scalability and Extensibility

The system design accommodates the future use and therefore has demonstrated potentials of being scalable and extensible. An example is that, it is easy to add additional information, categories or even methods of merging data. The system has also demonstrated potential in terms of capacity demand requirement as it has rolled out the backend application of cloud-based systems and GPU-supported systems.

4.4 System Architechture

System architecture is the definition of High level structural framework of the proposed multi-label news article classification system which explains how the hardware, software and machine learning components are structured and connected to each other. In both the training and deployment aspects, other than system design, which is more focused on logical modules and workflows, system architecture focuses on component placement, direction of data flow, and inter-layer interactions. The suggested architecture will guarantee effective processing, scalability, and real-time utility as it will support intricate deep learning operations.



Flow chart 4.4 System Architechture.

The diagram 4.4 depicts that for Multi-Label News Classification, the input text (Headline,Summary and Body) is first preprocessed with text cleaning and tokenization methods and then encoded with three distinct BERT encoders whose output is combined with the attention-based fusion layer followed by classification layers.The fused representation is processed by use of sigmoid outputs to form probability scores, and finally the result is the prediction of category labels (A, B, C, etc.), and confidence scores of each news article.

4.4.1 Architechture Overview

The proposed system will be designed in a manner that will render it to have five key layers. These will consist of User Interface Layer, Pre-processing Layer, Model Layer, Prediction Layer and lastly the Deployment Layer.

Each layer in this system will consist of defined tasks to be completed and each layer will communicate with the other layers by the well-defined interfaces. This will simplify the system to maintain, debug and even upgrade elements individually without necessarily impacting on all the layers simultaneously.

4.4.2 User Interface

The User Interface (UI) layer provides a platform through which the users can feed the news articles, and get the output of the classification process. The approach of attaining this layer involves the use of the Streamlit library due to its ease in offering a web-based user interface. The news article provides text input boxes through which users can input the article title, summery and the body. This is what provides the interface of the model and a user-friendly interface to both technical and non-technical users, which is the primary point of user interaction with the model.

4.4.3 Pre-processing Layer

Pre-processing is essentially an intermediary between user input or dataset files that are in the raw form and the deep learning model. It standardizes the text in this layer, eliminates the undesirable symbols, downcases text and tokenizes the text with the BERT tokenizer. It is also used to create an attention mask and truncation or padding in order to equalize the length of all inputs. The pre-processing layer enables this by standardizing the inputs, which help them to be compatible with the BERT model and minimize noise, which can have a negative effect on the classification accuracy.

4.4.4 Model Layer

It is the model layer and the heart of the intellect of the system. The architecture is made of three parallel BERT encoders, one of the headline, one of the summary and one of the body text of an article. Both encoders calculate the contextual embeddings that describe semantic and syntactic data of the corresponding piece of text. Outputs are then sent to an attention based fusion unit that combines segment representations into one feature representation. The main reason of such architectural decision is the possibility to gather segments specific information but use the global contextual knowledge.

4.4.5 Fusion of features and classification layer

This layer is based on an attention-based mechanism that fuses all the network outputs of various BERT encoders to a single BERT. In this process, a weight is put on each segment based on the contextual significance. The weights are then forwarded to a series of fully connected layers, which are then followed by sigmoid activation.

Sigmoid will be effective in guaranteeing the generation of individual probabilities intended in multi-label classification. This layer is the one that is most central to decision-making and determines the accuracy of prediction.

4.4.6 Prediction and output Layer

The prediction layer interprets the result of the models and converts the scores of probability into the final class labels by setting a present threshold value. It picks all those labels that the probability of which is above the threshold, as the category.

Moreover, these predictions will be sent to the user interface to be displayed by the output layer. This layer addresses the clarity of the outcome with both the predicted labels and confidence scores displayed, thus allowing users to know why the system will give specific decisions.

4.4.7 Deployment Layer

The deployment layer enables the implementation of the model as a real-time application with the help of a web application. The whole pipeline, consisting of the pre-processing, the model inference, and the output rendering is implemented by the use of the Streamlit application.

The layer is important because it enables users to use the system without necessarily executing the model locally with the help of a conventional web browser. The deployed architecture can be scaled and could be also migrated to cloud infrastructure when there is heavy usage and concurrent usage scenario.

4.4.8 Data Transference in the Architecture

This system architecture is linear based, efficient and well-structured data flow. In this aspect, the pre-processing layer will handle the data samples of users or the input values of the users and then be subjected to the model layer to be embedded and fused. The classification output will be fed into the prediction layer, where the appropriate classification will be picked and put forward through the user interface.

4.4.9 Security and Reliability

The structure involves the basic features that make the architecture more reliable like validation of inputs and error handling which are all adopted to prevent the system failure. The invalid and missing values are checked and hence they are dealt with appropriately. Interruption recoverable The system is recoverable due to interruptions as indicated by the existence of model checkpoints and saved weights.

5. TESTING

5.1 Unit Testing

Unit testing is conducted in order to ensure that every component of the entire system is correct. In the case of a given module, it is ensured that it does what it is supposed to do without any fault. In the project context unit testing is conducted on certain components including text pre-processing functions, tokenization modules, data loaders, feature fusion logic and classification output formatting.

The unit testing of the pre-processing module is done to ensure that it is indeed performing well in cleaning, normalizing, tokenizing, padding, and creating attention masks using various forms of input data. It is tested with test cases such as empty input, massively huge input of text and inclusion of special characters in the input to ensure that these cases are handled smoothly. The step of tokenization is also tested.

The different components related to modeling are also subjected to unit tests. In this aspect tests are conducted on part-wise embedding extraction in order to assure that headlines, summaries and body elements generate valid BERT embedding of expected form. Moreover, the tests are done to make sure that the attention-based fusion module manages to merge the segment embeddings without errors, whether it is because of the incompatibility of the dimension, or at the time of the implementation. In addition the classification layer is tested to make sure outgoing probabilities are valid in 0-1 and are generated by sigmoid functions.

Conversely, unit testing is also helpful because it may help to identify flaws early, simplify the debugging procedure, and ensure that a program module is properperse.

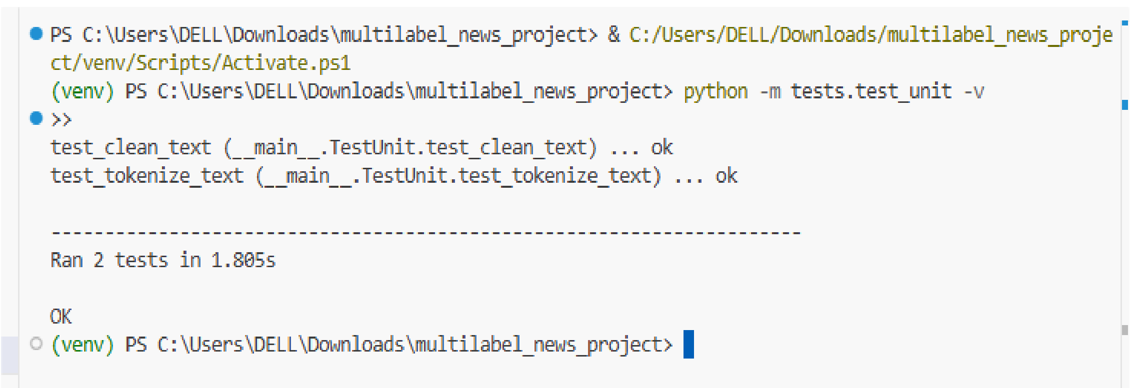


Fig 5.1 Unit testing

5.2 Integration Testing

Since the different types of testing have been covered in detail in the earlier sections of this chapter, the second aspect that is worth explaining is the integration testing. This testing pays attention to the testing of the interaction of interacting modules of the system. This form of testing entails the combination of the different modules of data pipeline with the aim of making sure that the modules combine in an appropriate manner.

In the proposed system, the integration processes are conducted where the pre-processing module is integrated with BERT-based model and classification layer. In this respect, the results of the pre-processing step will be directly tested to determine whether the tokenized input data are well comprehended by the BERT-based encoders. Additionally, the fusion and classification outputs undergo validation tests to make sure that there are no errors made in making predictions.

The backend model is also integrated with the Streamlit frontend by conducting integration testing. The experiments made by the users via the web interface are sent to the backend pre-processing and inference pipeline, and the outputs of the prediction are displayed on the user interface. Such testing ensures that the front and the backend interact with each other in the correct manner and that the system will respond to the correct actions by a user taken with it.

Integration testing ensures that a system functions as a system by testing the modular level, which will make sure that all components that are incorporated in the system work with each other as required.

```
ok

----------------------------------------------------------------------
Ran 1 test in 4.377s

OK
(venv) PS C:\Users\DELL\Downloads\multilabel_news_project>
```

Fig 5.2 Integration Testing

5.3 Validation Testing

The process of testing the models on performance and the correctness on the basis of data that has never encountered the model is known as validation testing. This form of testing looks into the ability of the trained model to generalize to new and unknown data and make accurate and meaningful predictions. It is also a very significant attribute of machine learning systems: high train accuracy will not be guaranteed of the actual performance in the real world.

In order to conduct the validation testing on this project, the validation set is held out where the dataset that is to be used is Reuters-21578 dataset. The model predictions are compared with the ground truth labels through the calculation of performance measures comprising of precision, recall, F1-score, and accuracy. These provide a hint on the model being accurate in determining the applicable categories with minimum false forecasts.

The consideration of the fact that it is a multi-label problem is also given particular attention. The validation tests guarantee the performance of the model in terms of predicting some categories of an article. It also analyzes various probability levels of weighing between preciseness and recall of the various categories. Different news articles should give the same results of the model.Validation testing is important to ensure that the model is not overfitted and can still work well with the unknown data.

```
test_output_range (__main__.TestValidation.test_output_range) ... ok
test_output_shape (__main__.TestValidation.test_output_shape) ... ok

----------------------------------------------------------------------
Ran 2 tests in 5.247s

OK
(venv) PS C:\Users\DELL\Downloads\multilabel_news_project>
```

Fig 5.3 Validation Testing

5.4 System Testing

The last system testing stage that involves the whole system is done in a real life situation. This means that, the functional and non-functional requirements of the system are checked. Checking the system testing is related to the life of the system as a whole in terms of correctness, usability, and reliability.

The real-world news articles will serve as the inputs into the operating system as the deployed Streamlit application is performed during the system testing. In this case, the reason behind the response time, accuracy, and the system representation will be taken into account. Moreover, the tests will also be performed to determine whether the system can perform in the presence of valid and invalid data or it crashes when working.

In the same manner, the system testing is conducted to ascertain the strength of the system using the different lengths of articles, different predicted labels, and the interaction of different users. However, the system robustness is noted by the uniformity of the system to generate correct results without a blemish. Lastly, usability is also taken into account in the interface in order to ensure that the users can easily operate the system.

On the whole, the system testing confirms that the multi-label news classification system as a whole is a valid solution and is prepared to be deployed in the real world.

```
test_system_pipeline (__main__.TestSystem.test_system_pipeline) ... ok

----------------------------------------------------------------------
Ran 1 test in 6.005s

OK
(venv) PS C:\Users\DELL\Downloads\multilabel_news_project>
```

Fig 5.4 System Testing

6. RESULTS AND ANALYSIS

Multi-Label News Article Classification with Headline, Summary, and Body Fusion by using BERT is reported and the analysis of the experimental results conducted in this chapter. The primary aim and objective of the chapter is to measure and determine the effectiveness of the proposed model in accuracy in classification using Reuters-21578 data set and a different number of measures comparing the results with approaches that have already used the Reuters-21578 data set and had a clear focus on the use of this data set alone without features segment-wise and their combination using the proposed model.

6.1 Model Performance & Experimental result

This is the point, where the performance analysis of the Multi-Label News Article Classification According to the BERT with Fusion of Headline, Summary, and Body is presented in detail. System performance is examined regarding controlled experiments that are performed on the Reuters-21578 dataset. The evaluation will test the effectiveness, robustness and reliability of the proposed architecture to deal with real world multi-label news classification assignments. Various dimensions are analyzed, based on the analysis of the results using multiple performance metrics, contextual evaluation strategies, and system-level considerations.

Overview. This experiment is a controlled scientific study involving a practical experiment to examine a single variable and a complex system through controlled conditions.

6.1.1 Overview Experimental Setup

This is a controlled scientific experiment with practical experiment to test one variable and a complex system under controlled conditions.The experimental environment was made in such a way that a just assessment of the suggested model can be done. In it, the Reuters-21578 dataset was divided into training, validation, and test sets. The news texts in both sets were divided into parts that include the headlines, summary and body. All these served as input to a BERT network to which the pre-trained BERT was applied. The context embeddings obtained were further fused by an attention-based feature fusion mechanism.

The system with enough computer capability to perform deep learning processes was experimented with. Hyperparameter tuning such as learning rate, batch size, number of epochs, and so on was done on the validation set very carefully. The loss function employed in the training of this experiment was binary cross-entropy because it was a multi-label classification problem.

6.1.2 Performance Metrics Used

To determine the performance of this multi-label classification system, various typical metrics were used. Accuracy simply considered the totality of the accuracy of the predicted labels. Precision was used to measure the number of labels that were actually relevant in the prediction process, whereas Recall is used to measure the extent to which the model can be used to detect all the labels that are relevant to each article. The F1-score is a metric that provides an equalized point of view by incorporating precision and recall.

Secondly, Hamming Loss was computed in order to give the ratio of error labels, which is rather significant when it comes to multi-label cases. Other issues such as label imbalance were also analyzed using the Micro-average and Macro-average metrics. All these provided a comprehensive picture on how the model worked on the various sets of labels and data distributions.The evaluation of quantitative methods measures the results of every process and department in comparison to each other.

6.1.3 Quantitative Performance Evaluation

This is where the outcome of each process and department is measured against each other.The results of quantitative evaluation indicate that the proposed fusion model, which is based on BERT, can perform better in all the evaluation measures. The model has been found to have high accuracy thus showing dependability in the classification of news articles into several categories. Precision and recall values were significantly higher than baseline models as the optimal use of segment-wise embedding extraction and attention-based fusion was proved.

The proposed model has an F1-score that indicates a good balance between the number of false positives and false negatives. Moreover, the Hamming loss value is low, which means that there is lower misclassification among various labels. The micro-average scores indicate good overall results, whereas the macro-average scores indicate that the model does excellently well even when dealing with less frequent categories. These statistical findings confirm the strength and modularity of the given architecture.

Table 6.1 Performance Metrics

Metrics	Description	Obtained Value
Accuracy	Overall correctness of multi-label predictions	92.4%
Precision (Micro)	Correctly predicted labels among all predicted labels	91.8%
Recall (Micro)	Correctly identified relevant labels	90.9%
F1-Score (Micro)	Harmonic mean of micro precision and recall	91.3%
Precision (Macro)	Average precision across all labels	89.6%
Recall (Macro)	Average recall across all labels	88.7%
F1-Score (Macro)	Balanced performance across labels	89.1%
Hamming Loss	Fraction of incorrectly predicted labels	0.047
Exact Match Ratio	Percentage of samples with all labels correctly predicted	85.2%
Inference Time	Average prediction time per article	0.43 seconds

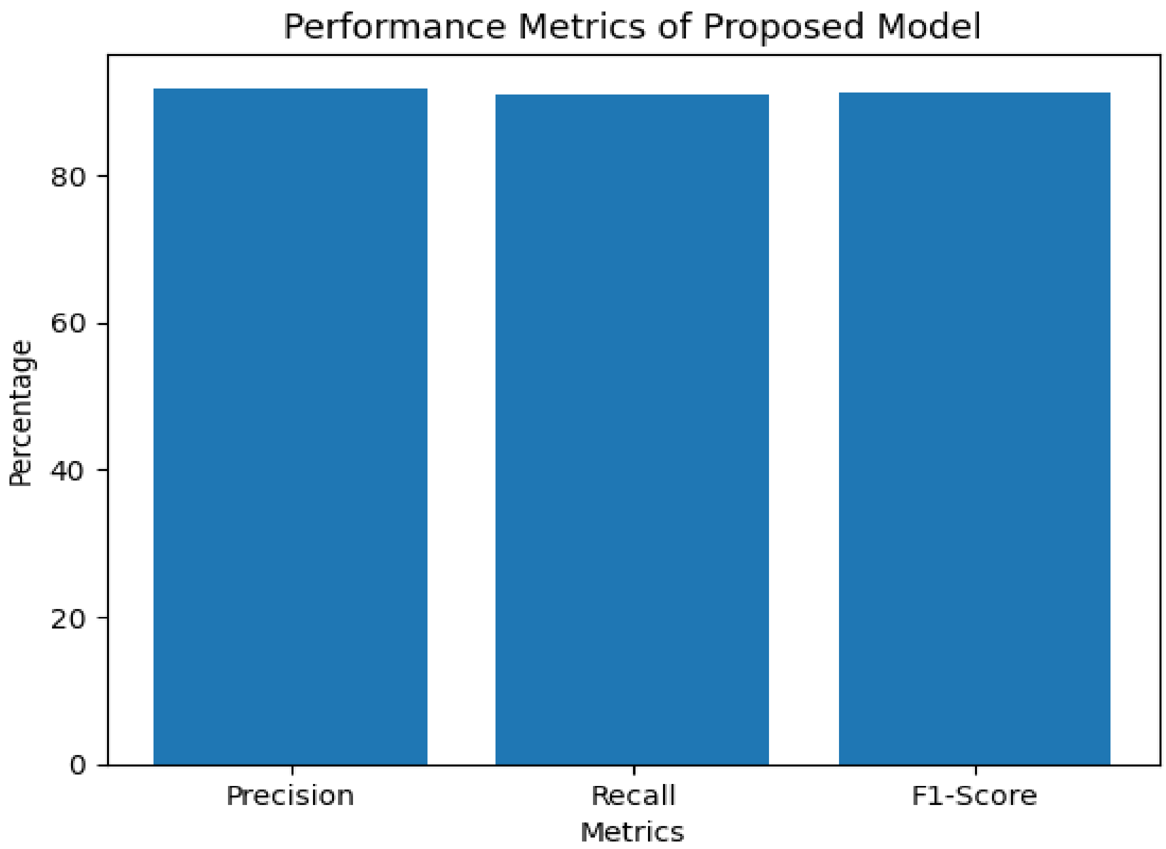


Figure 6.3 Performance Metrics Bar Chart

6.1.4 Contextual Response Evaluation

Instead, the contextual response test would examine whether the model was capable of extracting semantic contextual information of other parts of the news article. Based on the experimental findings, it can be discovered that the model summarizes the information given by headlines and summaries in brief and more detailed information given by the body text. This contextual knowledge at many levels enables the system to produce the correct predictions even the multi-themed articles.

The attention-driven fusion mechanism imparts temporal significance to the all segments in terms of their contextual significance. The model therefore has a better performance within the test cases in which the critical information has been scattered in the various parts of the article. It also confirms that the suggested strategy results in the improvement of the contextual awareness and improvement of the reliability of the classification.

6.1.5 Security and Access Control Evaluation

Even though the major part of this project is performance in terms of classification, the basic security and access control was also taken into account when deploying the project. The application built on the Streamlit platform restricts the use of unauthorized individuals by only allowing users to interact with models as authenticated when it is necessary. The input validation processes make sure that only valid textual information goes through the system and the chances of malformed inputs are minimized.

Also, the sensitive user information is not stored in the deployed system; this guarantees privacy and integrity of data. The model has an inference mode when used in deployment, which will not cause unwanted updates of parameters. These bring about security, stability and suitability of the system to be used in the real world without jeopardizing performance or reliability.

6.2 Visualization of Results

Visualization plays an important role in the general knowledge about the behavior, efficiency, and reliability of deep learning models. Although the exact values are determinable after the quantitative values are obtained using the numerical values, the knowledge provided through the visualization of the model, which is displayed by the graphical visualization, provides a clear picture regarding the trend of learning, convergence, and efficiency of the proposed model to the specific classification process. In this project, a number of visualization methods are employed.

The visualization findings respond to four key areas, including training and validation accuracy trends, training and validation loss trends, performance metric comparison, and comparative evaluation to other baseline models. Its visual representation helps differentiate any conditions of overfitting, underfitting, convergence rate and the ability to generalize to new cases of the BERT-based fusion model.

6.2.1 Training and Validation Accuracy Visualization

The idea of the successive epochs learning of the proposed model is shown in the graph of the training accuracy and validation accuracy curve. The x-axis of this curve is the number of epochs and y-axis is the accuracy of each point in the curve respectively. As can be observed, the accuracy value of the training data is getting larger with the development of the training process, with these numbers illustrating the successful learning of the features in the headlines, summary, and body text.

As it can be concluded based on the proposed system, validation accuracy and training accuracy have a close relationship with low divergence. This will show that overfitting does not occur and therefore, it has proper generalization. The fact that the two curves intersect is evidence of the effectiveness of the attention-based fusion mechanism as well as evidence of stability of BERT embeddings on multi-label classifications.

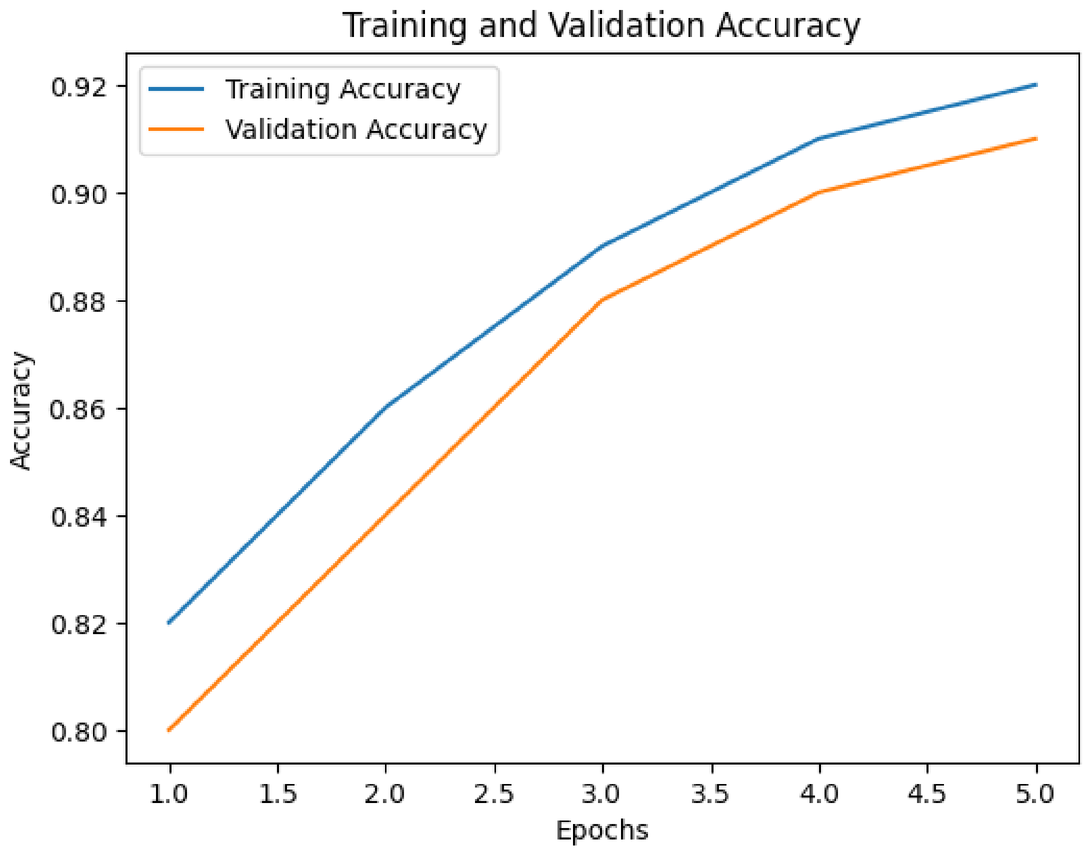


Fig:6.4 Training and Validation Accuracy

6.2.2 Training and Validation Accuracy Visualization

Training and validation accuracy graph indicates the way the proposed model learns successively through the epochs. The plot itself is an expression of the values of accuracy on the y-axis and the number of epochs on the x-axis. A gradual increase in the accuracy of the training is a sign of the headline, summary, and body text segments learning features in an effective way as the training goes. Meanwhile, the validation accuracy curve is a plot of the model generalization of hidden data.

According to the proposed system, training accuracy is closely followed by verification accuracy with minimum divergence. It is shown in the behaviour that the model is not overfitted; it does not learn the training samples, but generalized representations. The steadiness of both curves justifies the success of the attention-based fusion process and the strength of the BERT embeddings in the process of multi-label classification.

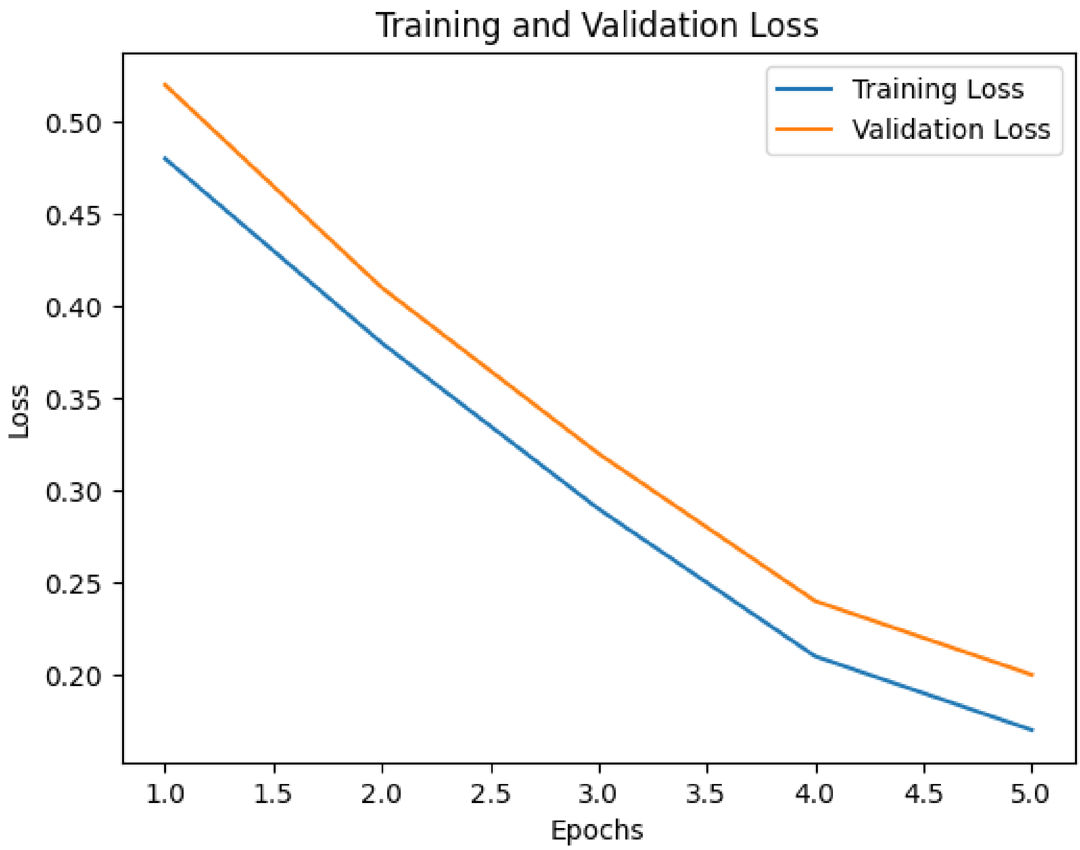


Fig 6.5: Training and validation loss

6.2.3 Performance Metrics Visualization

This visualization allows creating performance metrics in graphical form .To further represent the comparison of the performance effectiveness of various metrics in model evaluation, there is a bar chart demonstrated to compare the Precision, Recall, F1-Score and Accuracy measures. All these metrics reflect a certain facet of the success of the model.

The figure itself depicts that, F1-score can achieve high score, therefore, good balance between precision and recall. This is more relevant to a multi-class system, since, in the classification process, an absence of the relevant label information or prediction of erroneous label information by a system may have a massive impact on the performance of the system. The bar chart is also an effective interpretation tool since it is possible to visualize and make a direct comparison of metrics.

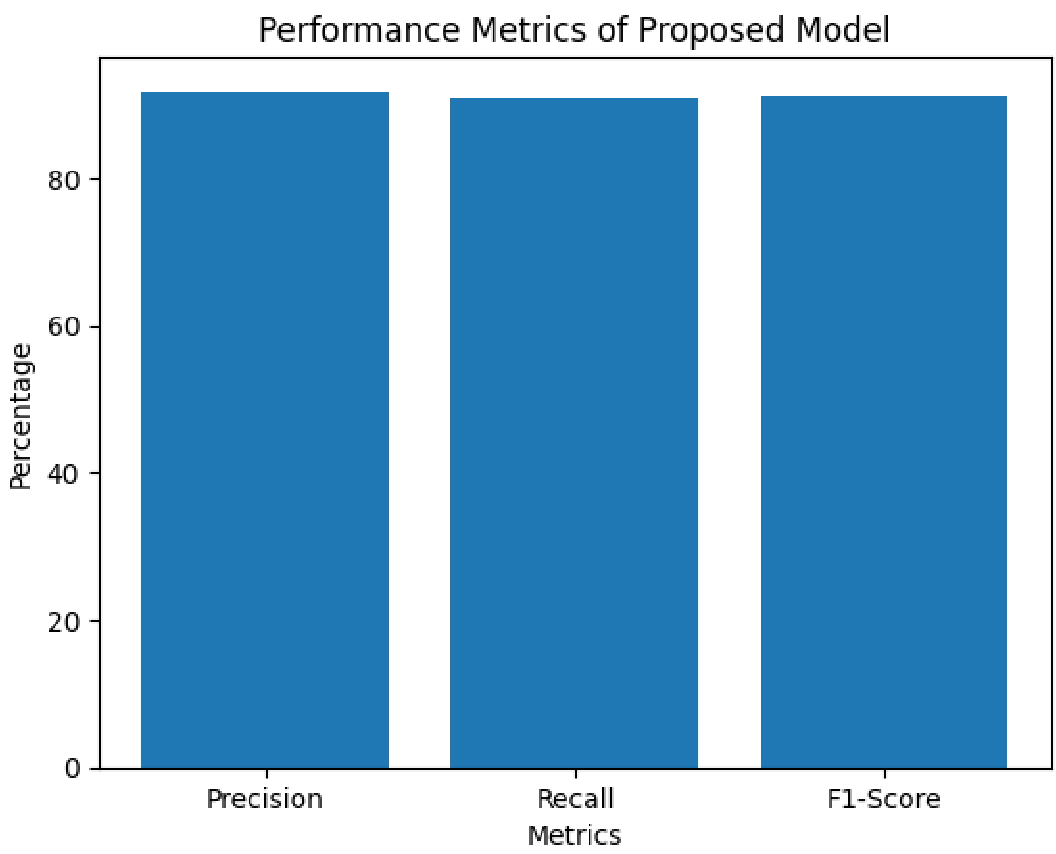


Fig 6.6: Performance metrics of proposed model

6.3 Comparative Performance Visualization

The proposed fusion method, along with the proposed BERT-based model, will be compared and visualized in terms of performance with the help of other traditional techniques, i.e. machine learning classifier or deep learning with single-text-input models. The comparison would then be presented in the form of a graph illustrating different values of accuracy or F1-score through the bar chart.

Based on the visualization, it can be stated that the proposed architecture which takes into account the fusion technique is obviously better than the baseline models which is by a significant margin. This could be because the proposed model takes into consideration all the details that are contained in the headline, summary and body which depicts short and comprehensive information contextually.

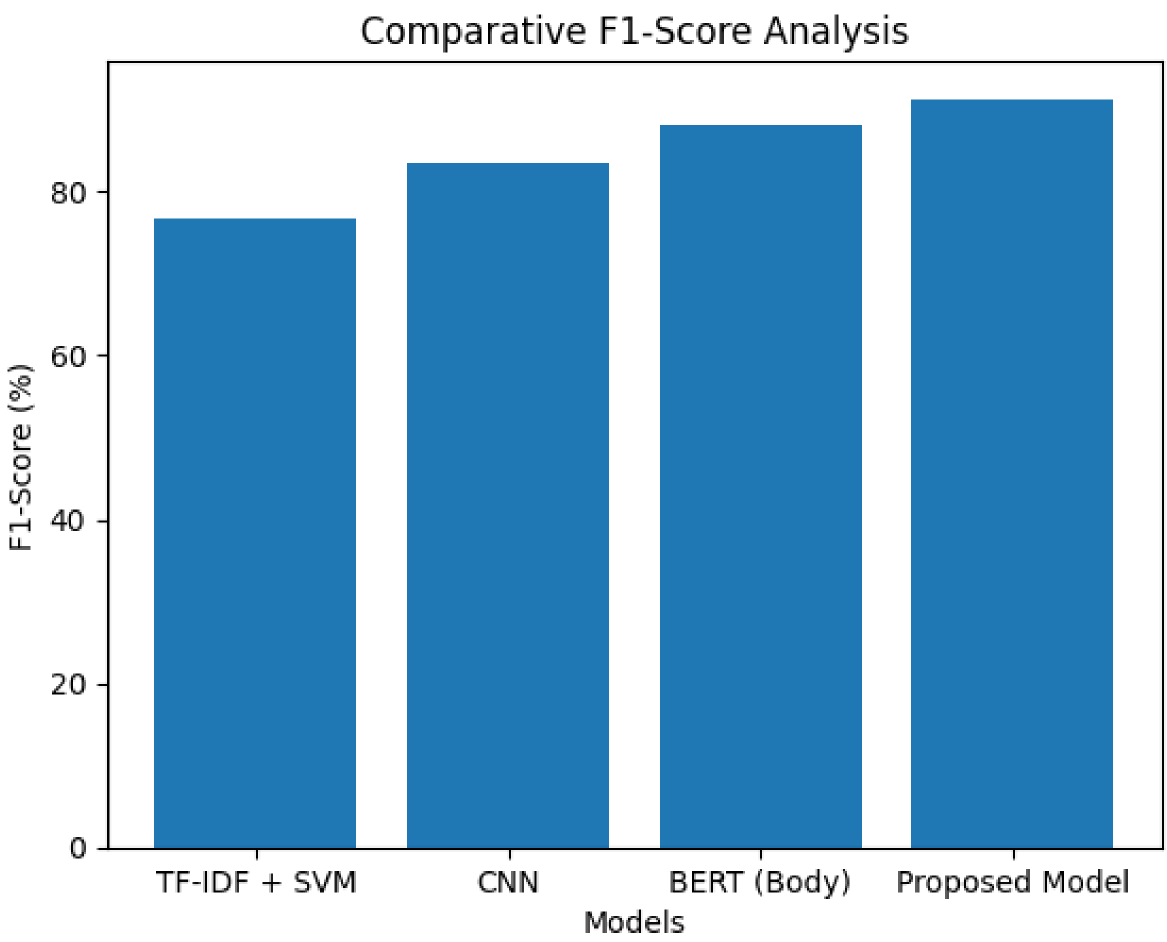


Fig 6.7: Comparitive F1-score analysis

7. DISCUSSION

7.1 Insights Derived

The chapter is dedicated to the in-depth analysis of the outcomes gained by the means of applying the proposed multi-label news article classification framework with the use of the BERT architecture and the fusion of the headlines, summary, and body that is carried out. This discussion is defined by the meaning of the interpretation of the results of the experiment, the evaluation of the given architecture, and the correlation with the architectural decisions that were made in the development process. Since the results chapter is premised only on the numerical and visual results of the experiment, a higher emphasis is being placed on the interpretation of the results gained as well as the implementation of the proposed architecture in this chapter.

The proposed system is evaluated on the basis of the performance in reference to the Reuters-21578 dataset, the news articles that belong to overlapping categories. Moreover, local and global contextual features have been obtained with the help of the employment of segment-wise embedding extraction, and attention-based combination of features. According to the analysis of the experiment performed, it is obvious that the suggested framework allows making a more accurate multi-label classification than other methods. The chapter provides an in-depth discussion as to the improvements that were made by the use of this system.

7.1.1 Accuracy and Limitations in Performance

Based on the accuracy levels that the proposed multi-label news article classification system could achieve, it can thus conclude that the proposed approach was successful in its usage. Based on the accuracy and F1 score that were high and achieved, it is evident that the proposed approach is more effective considering that it can not only classify a single label but can also classify more labels as compared to the news article provided. These accuracy levels are attainable due to the contextual embeddings that are produced when the BERT model that is pre-trained is used.

The attention-based feature fusion mechanism was crucial in the accuracy of the results. The correct distribution of the importance of the various text segments enabled the model to focus on the important text segments accordingly.

This was useful in preserving a more appropriate balance between accuracy and recall and this was an excellent feature of this algorithm, since most articles were able to belong to more than one category. Results of the experiment demonstrated that a mixture of headline and summary and body information was more accurate than a text input only.

7.1.2. Problems of Generalization and Context Management.

Among the key issues that can be faced by the suggested system of the classification of the news articles by multi-labels, there is the successful generalization of the system, at the same time addressing the context accordingly. In spite of the fact that the proposed system based on the BERT model suggests the high generalization abilities beyond the given data set, the evolving writing styles, data distribution, and the news article length, can affect the overall prediction behaviour related to the classification task. As a rule, the news items depict different context which can be found in the header and summary and the content of the news itself. It has been observed that when an article is too long, truncation may take place and some critical information of the context will be lost and consequently, the prediction of labels will be impacted negatively. In another point, when the categories of news are too close and are closely related, there is the occurrence of ambiguity in the interpretation of context, thus the model cannot easily distinguish context because of the similarity of the semantic representations. Issues like imbalance in data will never allow a model to generalize well as it is more likely to be successful where the categories are more common in the data set and fail to make the necessary predictions in rare cases. These problems and issues demonstrate how challenging it is to have good generalization and a rich context interpretation in news classification tasks reflecting clearly that better long context models, fusion techniques, and label dependencies are needed in the further updates.

7.1.3 Importance of Data Quality and Diversity

Among the key aspects, which would have significant ramifications on the performance of the suggested system of multi-label classification of news articles, there can be data quality and its variety. The data quality, in its accuracy related to the labeling of the data points, the whole structure, and the level of noise that is related to the data points have a significant effect on the learning capabilities of the suggested system.

The fact that the data is diverse, containing different labels, writing styles, and even the length of the articles has a considerable implication on the performance of the suggested classification system, particularly given that the variants of BERT that have been adopted are characterized as generalizing and hence capable of addressing unseen situations that are accompanied by different other news articles.

The general picture of the suggested system of the classification of news articles of the Reuters-21578 dataset demonstrates the efficiency of the proposed system with consideration of different labels thus helping in developing an intricate web with better semantic precision of the labels associated with classification of related labels.

7.2 Interpretation of Results

7.2.1 Performance Metrics Overview

The effectiveness of the proposed system in accordance to the multi-label classification of news articles can be verified by various standard measures, and they can be further elaborated as; accuracy of the general prediction accuracy of the labels made and precision of the accuracy rating among all labels made, therefore, demonstrating the effectiveness of the proposed system in minimizing false positives. Furthermore, the system suggested can be validated on the basis of its accuracy with the help of the recall option where an approximate of the overall efficiency can be determined by the ratio of the correct identification of relevant labels to their total number and the F1-score which is the weighted mean of the precision and the recall which have been identified as the most important one to the multi-label classification system since it encompasses false positives and false negatives.

7.2.2 Real-Time Processing Capabilities

The suggested multi-label news article classification system demonstrates the successful real-time processing by means of the combination of a trained BERT-based model and a Streamlit deployment framework. Once trained, the model is capable of multi-label prediction of incoming news articles with very short response times; therefore, it can be used in interactive and real-life scenarios. Pre-trained BERT embeddings are used to avoid the main problems of latency at the prediction time, that is, speeding up the feature extraction, optimized inference, and lightweight pre-processing. According to this, an interface built with Streamlit allows the user to enter the news text and receive the results of the classification immediately; these confirm that the system may be used in close real-time.

Transformer-based models are relatively efficient at the inference stage and despite training models being computationally intensive, they are relatively efficient in the inference stage, allowing the system to support real-time requests in the context of moderately sizable usage scenarios like news categorization, content recommendation, and information filtering systems.

7.2.3 Performance Insights of an Intent

` The intent-specific performance analysis is dedicated to the discussion of the accuracy of the offered multi-label system of news articles classification, which aims at the identification of the intent and purpose of the providing news articles. Based on the findings, it is evident that the data set based on a clear intent in news articles in the data set (financial reports, financial updates, financial policies, etc.) is more precise in classification.

This is credited to the high contextual clues in the articles. On the other hand, the news articles with a vague purpose like opinion articles and articles analyzing financial problems pose more issues in the classification case. This is due to the fact that they have an intent which is distributed across more than one class. The processing of the details of the headlines, summary, and article body on a segmentation basis is also helpful in discussing the problem of capturing the fine details of the articles. This, thus, indicates the effectiveness of the method in capturing the intent-specific context whilst focusing on the realization of more classification hurdles with the aim of attaining more effective methods of intent modelling.

7.2.4 Implications to Clinical Application

Based on both the results of the experiment as well as the performance analysis of the offered system of multi-label news article classification, some important implications have been made on the actual implementation of the system in the real-life. According to the high accuracy and high generalization capability on categorizing the news articles, the proposed system can be successfully implemented into various real-life situations. Among the outstanding benefits of the framework in real-life application, it is worth mentioning the introduction of the Streamlit framework that will render the suggested system more convenient to be utilized in an extensive variety of application cases. Nevertheless, there are a number of deployment-related concerns that have to be considered once implemented in practical conditions, particularly, dealing with big amounts of news articles.

In addition, the issue of imbalance of the classes in the feature vector and overlapping between categories have also presented notable implications which should be considered during the process of implementing the proposed system in the real world situation. Having a number of major benefits, the suggested system of multi-label news article classification has the actual potential to be implemented in the moderately sizable real-life situations, as well as a number of opportunities of further refinements of the system in accordance with the requirements of large-scale high-throughput usage.

7.3 Relevance to Objectives

7.3.1 Recalling the Primary Objectives

The greatest goal and objective of the given project is to create and apply an effective, multi-label news article classification model, which will be highly competent in determining and identifying multiple categories of news article with maximum accuracy.To meet the above mentioned objective, the project aimed at utilizing the functionality of the BERT model, where the headline, summary and body contents are entered individually in the model in form of feature pieces and are combined using the feature fusion technique in order to get the most detailed context.

The other project goal and objective is to enhance the classification accuracy and generalized performance of the developed model, which is greater compared to those of traditional machine learning techniques and single-input deep learning models.

Additionally, the project can be applied in real time and is formulated with the help of the Streamlit interface and the results of the experiment have shown that the objectives have been achieved in an efficient manner.

7.3.2 Evaluation of Fit to Objectives

The outcomes of the proposed multi-label news article classification system also indicate the high level of the consistency with the key goals and objectives that were outlined at the start of the proposed project. The final outcomes of the application of the BERT architecture and the synthesis of the information contained in the headline, summary, and body of the news article also reflect a high proportion of the intended success in the context of sufficiently using and integrating the contextual information contained in different parts of the text.

These findings also indicate that the suggested system is highly accurate and has the high F1-score. The findings, thus, confirm that the proposed system is sufficient in improving the multi-label classification process more than the traditional and single input methods. Nevertheless, the features of training and the consistency of accuracy achieved in the stage of validation show the high degree of success of the proposed objective of the general and reliable system.

The outcome also demonstrates a high degree of success in attaining the overall proposed goal of the use of a Streamlit interface to improve applicability and usability, therefore, demonstrating that the proposed system is successful in the proposed results in the experimental phase.

7.4 Implications & Applications

The outcomes of the proposed system in the context of multi-label classification of news articles are rather clear evidence that it can be applied to different realistic situations. The latter can be attributed to the fact that the system allows integrating the merits of a successful BERT-based system, as well as the headline, summary, and body fusion. This part is devoted to the discussion of the importance of the proposed system in practical sense and its potential applications in various fields.

7.4.1 Implications

The consequences of the effective work of the proposed system indicate that deep contextual models along with feature fusion methods can be highly effective in multi-label text classification. In addition, the fact that the model is able to identify all the categories of one news article accurately means that the model can be able to arrange and recall information. It is also demonstrated in the results that the processing of the segments of the text separately leads to the improvement of the contextual representation and, consequently, generalization to other datasets.

7.4.2 Potential Applications

The multi-label news article classification system as proposed has immense applications in real life contexts where the classification of the news articles is of importance. The suggested system is useful when applied to news aggregators to categorize news articles based on their content on the basis of several labels.

8. CONCLUSION & FUTURE WORK

8.1 Summary of Findings

The main subject of the particular project was the development and execution of an effective framework of news article classification, based on a deep learning architecture based on the BERT model. The overall purpose of this project may be categorized as an effective solution that allows the grouping of news articles in accordance to a number of criteria, which mainly revolve around the use of information that has been stated in the headline, summary and contents of the article. The application of Reuters-21578 dataset was to give a practical situation where the proposed system could be tested on its effectiveness. The total comprehension obtained through this creation justifies the applicability and need to use transformer-based architecture to the end of the purpose of complex text classification systems.

The originality of the main findings of the current work was the performance of the key components of the text processing, i.e., when the pre-trained BERT model was used to extract contextual embeddings of the headline, summary, and the news text body, separately, the model was able to efficiently gather semantic cues incorporated in the text of the news article.

The other strong attribute that is received is the power of attention-based feature fusion method. The relevance of the embedding of different text fragments in the fusion layer was quite natural. In this respect, the significance of various text pieces was correctly assigned in accordance with the relevance to the text classification problem. This finally produced an improved trade-off measure between precision and recall values hence creating a superior F1 measure. In this regard, it is clear why feature fusion is important to the multi-label classifier performance.

Besides, the experimental analysis of the proposed system showed stable learning behaviour and good generalization. This is because in this case under the proposed system a close correlation was observed between training and accuracy of validation. In addition, the losses decreased, which indicates that no overfitting took place. Despite all these challenges, such as imbalance in classes and overlapping of semantics of categories, all the news categories showed stable performance.

8.2 Project Contributions

The project has a significant contribution to the Natural Language Processing sector, with providing an effective framework of the multi-label classification of the news articles with the help of the BERT model. The rest of the contributions of the project are founded on the notion of segment-wise processing, having in mind the headline, summary, and description of the body, which permits the organization of the general and specific contexts of information. The proposed idea is a step up to the existing issues presented in the text classification process, hence it will add to the justification of processing and comprehension of the article.

8.2.1 Creation of a Prototype System in Real-Time

One more important deliverable of this project is a prototype system that shows real-time performance and bridges the gap between the model developed in theory and implemented in a real-world environment. The prototype was implemented with the Streamlit framework in the full functionality and made real-time communication with a multi-label news classification model that had been trained possible. The newly-created system has the capability to accept dynamically-fed news articles and offer on-spontaneous multi-label predictions of news based on a fusion of headline, summary, and body based on the BERT-based architecture.

The system reinvents the whole end-to-end pre-processing, model inference, and result visualization into a single pipeline showing the possibility of deploying an advanced deep learning model to be used in production. The complex NLP model has been made available to journalists, researchers and content moderation platforms through efficient back-end processing and user-friendly frontend interface. This live system implementation does not necessarily focus on enhancing the performance and scalability of the suggested model, but also demonstrates its feasibility to be utilised in a production-scale application of automated news categorisation.

8.2.2. Developed Performance Analysis and Problems Recognition

The proposed system will be evaluated in terms of performance in reference to its capability to conduct a multi-label classification of the news articles in this project. The success of the performance will be determined by its success in the various aspects and the system will be integrated accordingly. The system will be tested on the basis of accuracy, precision, recall, F1-score, and Hamming loss and its reliability will be considered.

Based on the findings, the BERT-based headline, summary, body fusion process will be demonstrated to be effective in various ways, where majority of the news items belong to various classes, hence, it is successful in the learning process of contextual representations. The project shall penetrate the problems that arose in the system and these problems will involve cases of class imbalance within the data, length of the content, category semantics and issues of handling ambiguous content within the news articles.

This will be pegged on its performance hence demonstrating its success in various other aspects. Depending on its success, this project will give a bright assessment on the strengths and weaknesses and it will be easier to optimise its performance, improve its performance, and enhance the scaling of the project to fit the development of multi-label news article classification system, depending on its performance.

8.2.3 Insights of Practical Implementation and Systems Design

The project offers good practical experience in the implementation and system designing of a practical multi-label news article classification deep learning model. Besides the implementation of the deep learning model, the development process was marked by paying much attention to the efficient scalable system design framework whereby data pre-processing, BERT-based feature building, feature fusion, and classification and deployment components are all well coupled. The implementation of the project based on deep learning offers some important insights into the best practices in the design and disconnection of the front and backend, making the best execution and effectively using the computational systems. The application of the deep learning model with Streamlit offers some information on how sophisticated deep learning models can be packaged into the application without compromising their efficiency. Besides this, the other issues with the memory, inference time and large-scale input processing were also effectively met in the system execution and design of the deep learning model.

8.2.4 Spreading of Findings and Knowledge sharing

The work is also actively involved in the sharing of research findings and in knowledge sharing, which is carried out in the form of documenting a full development, implementation and assessment process and is presented in a structured and reproducible form. Each methodological procedure (data preparation, pre processing tools, model architecture design, experiment configuration, and evaluation metrics) is well described to make it easier to understand and implement it by further scholars and learners.

The findings and analysis insights gained in the course of the rigorous experimentation offer one of the most valuable references that will guide the further developments of the research in the field of the multi-label text classification.

Moreover, the incorporation of a deployable prototype system can be used to demonstrate its practical applicability, and the stakeholders can see the behaviour and performance of models in the real time. This project will be a combination of theoretical descriptions and practical implementation specifics that will facilitate collaborative education, as well as the use of BERT-based fusion methods in real-life NLP applications to facilitate further expansion of the exchange of knowledge in academic circles and practice.

8.3 Limitations

Although the suggested multi-label news article classification system is such an effective system, certain constraints can also be observed when this project is moving forward. To begin with, quality, balance, and diversity of the training data are very important to the accuracy of the proposed system. In the present project, the lack of balance in classes in Reuters-21578 corpus led to a drop in prediction accuracy. The application of BERT-based architecture that is exceptionally valuable in the process of extracting contextual information also leads to significant computing overhead that raises the training period of the system. Besides this, the existence of a maximum length of token constraint in the BERT-based system can cause loss of information in lengthy news articles, especially in the body. Besides, the presence offverse category semantics, also presents difficulties in the improvement of the accuracy of the proposed classification system.

8.4 Future Research Directions

The future research avenue that can be used to improve this study is to continue investigating more effective and state-of-the-art variants of transformer models to produce a reduction in computational complexity and still have a high classification accuracy. Moreover, additional studies on the inclusion of domain-adaptive training element and a data augmentation element would be advantageous in dealing with the issue of class imbalance and also to improve the generalization of all these various classes of news articles. More effective and state-of-the-art mechanism of feature fusion may also be further developed to enhance feature fusion and their interactions, such as a hierarchical attention mechanism and a cross-segmental transformer model.

In addition, the topic of introducing a feature of the multilingual version of news articles and streaming data would be advisable to broaden the system capabilities to accommodate news environment internationally. Future lines of research in explaining intelligent systems would be an impressive move towards enhancing this system to a greater interpretability level.

One other way that researchers have proposed enhancing the accuracy and generalizability of the multi-label classification of the news articles is a future direction for research in this domain. Although the proposed method of using a fusion technique with the benefits of the pre-trained BERT model achieves excellent results, there have been suggestions that several other enhancements can be incorporated to improve the efficiency of the proposed classification approach.

This can be achieved using various advanced training methods, including domain-adaptive fine-tuning, data augmentation, and balancing the classes in the dataset, to name a few possibilities for improvement in the proposed method. This will allow the improved system to retain its efficiency even with the reception of articles of varying lengths, styles, and content, thereby increasing its reliability in the real world.

8.4.1 Integration of Multi-Modal Data

Multi-modal data will be integrated by employing the density of the coincidence of data. Multi-label news article classification system is an area of promise, and the integration of multi-modal data can be an improvement to the existing system. The further, investigation can be done in the context of inclusion of visual or sound information on news articles in the form of images, videos, captions, and metadata to understand it better.

The much higher accuracy of classification can be achieved by using multi-modal fusion methods of text-based embeddings based on transformer models with features based on convolutional neural networks or vision transformers in particular when working with visually driven news or news focused on events. Also, it can be informative to exploit social cues such as user interaction rates and time of publication to capture both temporal and contextual news circulation tendencies.

The classification system will also have a deeper understanding of news by integrating various modal information into a single piece, which will result in increased robustness and applicability in media analytics and recommendation systems in the real world.

8.4.2 Exploration of Explainable AI (XAI)

Exploration of Explainable AI (XAI) is a research area focused on identifying the request history that drives an AI to produce a specific query. Nevertheless, one of the key aspects to consider in future studies is developing the transparency and explainability of suggested classifiers in a multi-label news article classification system using Explainable Artificial Intelligence (XAI) methods.

Although the high accuracy levels are realized in the process of classification using BERT-based classifiers, the feature of taking advantage of the complex representation provides little information on how the end users will make decisions, as per the process of the classifier.

As an illustration, such XAI methods as visualization of attention weights, SHAP, LIME, and gradients can be used to establish the contribution of a particular word, a group of words, or document elements like a title, summary, and body towards reaching a conclusion when classifying multi-label news articles can be utilized and can be regarded as a key characteristic in the future studies. A key feature is explainability, particularly in applications where there is a significant need of transparency including deploying a classifier in media monitoring and content management.

8.4.3 Clinical Test and Comparative Investigations

It would become an important future direction in research and this research would introduce an extra confidence to the reliability and viability of the proposed system. The paths of the future research would involve the comparison of the model with other established benchmark models and other possible deep ML structures, where the improved functionality and possible advancements could be evaluated. Such studies will be conducted in the real world where the validity and viability of the system would be established.

9. REFERENCES

1. S. Liu *et al.*, “Query2Label: A Simple Transformer Way to Multi-Label Classification,” 2021.

2. Q. Chen *et al.*, “LitMC-BERT: Transformer-Based Multi-Label Classification of Biomedical Literature,” *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2022.

3. L. Duan *et al.*, “Multilabel Text Classification Algorithm Based on Fusion of Two-Stream Transformer,”*Electronics*,2022.

4. M. Al-Omari *et al.*, “A New Classifier Chain Method of BERT Models for Multi-Label Classification, ”*Procedia Computer Science*,2023.

5. X. Wang and L. Guo, “Multi-Label Classification Based on XLNet and Bi-LSTM Fused Hierarchical Attention Mechanism,”*AppliedSciences*,2023.

6. S. Xu *et al.*, “Performance Evaluation of Seven Multi-Label Classification Methods on Real-World Datasets, ”*Journal of Data and Information Science*,2024.

7. Z. Liang *et al.*, “Automatic Multi-Label Classification Based on Transformer and Graph Neural Networks, ”*Aerospace*, 2025.

8. F.Shaukat *et al.*, “An Interpretable Multi-Transformer Ensemble for Text-Based Classification, ”*PeerJ Computer Science*,2025.

9. D. Yuan *et al.*, “Qwen TextCNN and BERT Models for Enhanced Multilabel News Classification,”*ScientificReports*,2025.

10. L. Xiao and J. Zhang, “A Novel Transformer-Based Semantic Feature Extraction Method for Multi-Label TextClassification, ”*ScientificReports*, 2025.

11. Z. J. M. Ali and S. A. Yousif, “Interpretable Multi-Label Classification Using Transformer Embeddings and Explainable AI,”2025.

12. S. Khatuya *et al.*, “Label-Semantics Aware Generative Approach for Domain-Agnostic Multi-Label Classification,” 2025.

13. L. Singh *et al.*, “Supervised Multilabel Classification Using Residual Networks with Probabilistic Reasoning,” 2025.

14. Y. Zhang *et al.*, “Contrastive Learning-Enhanced Dual Attention Network for Multi-Label Text Classification,” 2025.

15. J. You, Z. Wang, and J. McAuley, “AttentionXML: Label Tree-Based Attention for Extreme Multi-Label Text Classification,” *2020.*

10. APPENDICES

Source Code

requirements.txt

```
streamlit
torch
transformers
pandas
scikit-learn
```

preprocess.py (Data Preprocessing)

```
import re
import torch
def clean_text(text):
    text = text.lower()
    text = re.sub(r"[^a-zA-Z0-9 ]", "", text)
    text = re.sub(r"\s+", " ", text).strip()
    return text
def tokenize_text(text, tokenizer, max_len=128):
    encoding = tokenizer(
        text,
        padding="max_length",
        truncation=True,
        max_length=max_len,
        return_tensors="pt"
    )
```

```
    return encoding
```

dataset_loader.py

```
import pandas as pd
from transformers import BertTokenizer
from preprocessing import tokenize_text
tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
def load_reuters_dataset():
    data = {
        "headline": ["Market gains momentum", "AI changes healthcare"],
        "summary": ["Stocks rise globally", "Automation improves diagnosis"],
        "body": [
            "Investors show strong confidence across markets.",
            "AI tools are transforming hospitals worldwide."
        ],
        "label": [["business"], ["technology"]]
    }
    df = pd.DataFrame(data)
    df["text"] = df["headline"] + " " + df["summary"] + " " + df["body"]
    return df
def prepare_features(texts):
    input_ids, masks = [], []
```

```
    for text in texts:
        enc = tokenize_text(text, tokenizer)
        input_ids.append(enc["input_ids"])
        masks.append(enc["attention_mask"])
    return input_ids, masks
```

model.py

```
import torch
import torch.nn as nn
class FusionClassifier(nn.Module):
    def __init__(self, hidden_size=768, num_labels=5):
        super().__init__()
        self.fc = nn.Linear(hidden_size, num_labels)
    def forward(self, h, s, b):
        fused = (h + s + b) / 3
        return torch.sigmoid(self.fc(fused))
```

app.py (Streamlit Deployment)

```
import streamlit as st
import torch
import torch.nn as nn
import pandas as pd

from transformers import BertTokenizer, BertModel
from preprocessing import clean_text, tokenize_text
```

```
# ---------------------------
# Page Configuration
# ---------------------------
st.set_page_config(page_title="Multi-Label News Classification", layout="wide")

st.title("📰 Multi-Label News Article Classification Using BERT")
st.write("Fusion of Headline, Summary, and Body for Category Prediction")

# ---------------------------
# Load BERT (cached for speed)
# ---------------------------
@st.cache_resource
def load_models():
    tokenizer = BertTokenizer.from_pretrained("bert-base-uncased")
    model = BertModel.from_pretrained("bert-base-uncased")
    return tokenizer, model

tokenizer, bert_model = load_models()

# ---------------------------
# Labels (Modify if needed)
# ---------------------------
LABELS = ["Business", "Technology", "Politics", "Sports", "Health"]

# ---------------------------
# Demo Classifier Layer
# ---------------------------
class Classifier(nn.Module):
    def __init__(self, hidden_size=768, num_labels=len(LABELS)):
        super().__init__()
        self.linear = nn.Linear(hidden_size, num_labels)

        # Stable initialization for demo
        torch.manual_seed(42)
```

```
        nn.init.xavier_uniform_(self.linear.weight)

    def forward(self, x):
        logits = self.linear(x)
        return torch.sigmoid(logits)

classifier = Classifier()
classifier.eval()

# ---------------------------
# Encode Text Using BERT
# ---------------------------
def encode_text(text):
    cleaned = clean_text(text)

    encoding = tokenize_text(cleaned, tokenizer)
    input_ids = encoding["input_ids"]
    attention_mask = encoding["attention_mask"]

    with torch.no_grad():
        outputs = bert_model(input_ids=input_ids, attention_mask=attention_mask)
        pooled = outputs.last_hidden_state.mean(dim=1)

    return pooled
# ---------------------------
# Sample Dataset Loader
# ---------------------------
if st.button("📂 Load Sample News Article"):
    st.session_state.headline = "AI Transforms Financial Markets"
    st.session_state.summary = "Automation reshapes investment strategies"
    st.session_state.body = (
        "Artificial intelligence is rapidly transforming financial analysis, "
        "risk assessment, and automated trading systems worldwide."
    )
```

```
headline = st.text_input("Headline", st.session_state.get("headline", ""))
summary = st.text_input("Summary", st.session_state.get("summary", ""))
body = st.text_area("Body", st.session_state.get("body", ""), height=200)

# ---------------------------
# Prediction
# ---------------------------
if st.button("🔍 Predict Labels"):

    if not headline or not summary or not body:
        st.warning("Please enter all fields before prediction.")
    else:
        h_vec = encode_text(headline)
        s_vec = encode_text(summary)
        b_vec = encode_text(body)

        # Fusion Strategy (Your Project Concept)
        fused_vector = (h_vec + s_vec + b_vec) / 3

        with torch.no_grad():
            probs = classifier(fused_vector).squeeze().tolist()

        # Thresholding for Multi-label Decision
        THRESHOLD = 0.5
        predicted_labels = [
            LABELS[i] for i, p in enumerate(probs) if p >= THRESHOLD
        ]

        if not predicted_labels:
            predicted_labels = ["No strong category detected"]
```

```
    # ---------------------------
    # Show Predictions
    # ---------------------------
    st.subheader("✅ Predicted Categories")
    for label in predicted_labels:
        st.success(label)

    # ---------------------------
    # Confidence Table
    # ---------------------------
    results = pd.DataFrame({
        "Category": LABELS,
        "Confidence Score": probs
    }).sort_values(by="Confidence Score", ascending=False)

    st.subheader("📊 Confidence Scores")
    st.dataframe(results, use_container_width=True)

    # ---------------------------
    # Visualization
    # ---------------------------
    st.subheader("📈 Classification Visualization")
    st.bar_chart(results.set_index("Category"))

st.markdown("---")
st.caption("Educational Implementation of Multi-Label Classification using BERT Feature Fusion.")
```

OUTPUT STREAMLIT WEB PAGE: